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Electromagnetic Waves

JEE Main Physics · Previous Year Questions with Answer Key

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MCQ (Single Correct Answer) 163 questions

Q1. JEE Main 2026 (Online) 6th April Evening Shift

For an electromagnetic wave propagating through vacuum, $\vec{k}$, $\vec{E}$ and ω represent propagation vector, electric field and angular frequency, respectively. The magnetic field associated with this wave is represented by:

Open solution

(A) $\frac{\vec{E}\times\vec{k}}{\omega}$ (B) $\frac{\vec{k}\times\vec{E}}{\omega}$

(C) $\omega(\vec{E}\times\vec{k})$ (D) $\omega(\vec{k}\times\vec{E})$

Q2. JEE Main 2026 (Online) 6th April Morning Shift

A point light source emits E.M. waves in free space. A detector, placed at a distance of $L\,\mathrm{m}$, measures the intensity as I_o. The detector is now shifted to another location on the same spherical surface ensuring the angle between original location and new location as 45°. The measured intensity at new location will be ____ .

Open solution

(A) $\frac{I_o}{4}$ (B) I_o (C) $\frac{I_0}{\sqrt{2}}$ (D) $\frac{I_o}{2}$

Q3. JEE Main 2026 (Online) 5th April Evening Shift

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : The electromagnetic wave exerts pressure on the surface on which they are allowed to fall.

Reason (R) : There is no mass associated with the electromagnetic waves.

In the light of the above statements, choose the correct answer from the options given below :

Open solution

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)

(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(C) (A) is true but (R) is false

(D) (A) is false but (R) is true

Q4. JEE Main 2026 (Online) 5th April Morning Shift

A displacement current of 4.0 A can be set up in the space between two parallel plates of $6\mu\,\mathrm{F}$ capacitor. The rate of change of potential difference across the plates of the capacitor is nearly $\alpha\times10^6\,\mathrm{V/s}$. The value of α is ____ .

Open solution

(A) 0.58 (B) 0.67 (C) 0.82 (D) 0.75

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Q5. JEE Main 2026 (Online) 4th April Evening Shift

A magnetic field vector in an electromagnetic wave is represented by $\vec{B} = B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{j}$. Its associated electric field vector is ____ .

(A) $\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

(B) $\vec{E} = -v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{i}$

(C) $\vec{E} = v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{k}$

(D) $\vec{E} = v\lambda B_0 \sin\left(2\pi vt - \frac{2\pi x}{\lambda}\right)\hat{i}$

Open solution

Q6. JEE Main 2026 (Online) 2nd April Evening Shift

An electromagnetic wave travels in free space along the *x*-direction. At a particular point in space and time, $\vec{B} = 2 \times 10^{-7}\hat{j}$ T is associated with this wave. The value of corresponding electric field $\vec{E}$ at this point is _____ V/m.

(A) $60\ \hat{k}$ (B) $-60\ \hat{k}$ (C) $30\ \hat{k}$ (D) $-600\ \hat{k}$

Open solution

Q7. JEE Main 2026 (Online) 2nd April Morning Shift

An electromagnetic wave travelling in **x-direction** is described by field equation

$E_y = 300 \sin\omega\left(t - \frac{x}{c}\right)$.

If the electron is restricted to move in **y-direction** only with speed of 1.5×10^6 m/s then ratio of maximum electric and magnetic forces acting on the electron is _____.

(A) 200 (B) 150 (C) 400 (D) 300

Open solution

Q8. JEE Main 2026 (Online) 28th January Evening Shift

A plane electromagnetic wave is moving in free space with velocity $c = 3 \times 10^8$ m/s and its electric field is given as $\vec{E} = 54\sin(kz - \omega t)\,\hat{j}$ V/m, where $\hat{j}$ is the unit vector along y-axis. The magnetic field vector $\vec{B}$ of the wave is :

(A) $-1.8 \times 10^{-7}\sin(kz - \omega t)\,\hat{i}$ T

(B) $+1.8 \times 10^{-7}\sin(kz - \omega t)\,\hat{i}$ T

(C) $1.4 \times 10^{-7}\sin(kz - \omega t)\,\hat{k}$ T

(D) $1.4 \times 10^{-7}\sin(kz - \omega t)\,\hat{i}$ T

Open solution

Q9. JEE Main 2026 (Online) 28th January Morning Shift

The electric field of an electromagnetic wave travelling through a medium is given by $\vec{E}(x,t) = 25\sin\left(2.0 \times 10^{15}t - 10^7 x\right)\hat{n}$ then the refractive index of the medium is ____ .

(All given measurement are in SI units)

(A) 2 (B) 1.2 (C) 1.5 (D) 1.7

Open solution

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Q10. JEE Main 2026 (Online) 24th January Morning Shift

Open solution

Match the LIST-I with LIST-II

List-I		List-II	
A.	Radio-wave	I.	is produced by Magnetron valve
B.	Micro-wave	II.	due to change in the vibrational modes of atoms
C.	Infrared-wave	III.	due to inner shell electrons moving from higher energy level to lower energy level
D.	X-ray	IV.	due to rapid acceleration of electrons

Choose the correct answer from the options given below:

(A) A-IV, B-II, C-I, D-III

(B) A-IV, B-III, C-I, D-II

(C) A-IV, B-I, C-II, D-III

(D) A-II, B-IV, C-III, D-I

Q11. JEE Main 2026 (Online) 23rd January Evening Shift

Open solution

The ratio of speeds of electromagnetic waves in vacuum and a medium, having dielectric constant $k = 3$ and permeability of $\mu = 2\mu_0$, is (μ_0 = permeability of vacuum)

(A) $6 : 1$

(B) $3 : 2$

(C) $\sqrt{6} : 1$

(D) $36 : 1$

Q12. JEE Main 2026 (Online) 23rd January Morning Shift

Open solution

Match List - I with List - II.

List - I Relation		**List - II Law**	
A.	$\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{d}{dt} \oint \vec{B} \cdot \overrightarrow{da}$	I.	Ampere's circuital law
B.	$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 \left(I + \epsilon_0 \frac{d\phi_E}{dt} \right)$	II.	Faraday's laws of electromagnetic induction
C.	$\oint \vec{E} \cdot \overrightarrow{da} = \frac{1}{\epsilon_0} \int_v \rho \mathrm{dv}$	III.	Ampere - Maxwell law
D.	$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 I$	IV.	Gauss's law of electrostatics

Choose the correct answer from the options given below :

(A) A-I, B-IV, C-III, D-II

(B) A-II, B-III, C-IV, D-I

(C) A-IV, B-I, C-II, D-III

(D) A-II, B-III, C-I, D-IV

Q13. JEE Main 2026 (Online) 22nd January Evening Shift

Open solution

A laser beam has intensity of $4.0 \times 10^{14}\ \mathrm{W/m^2}$. The amplitude of magnetic field associated with beam is ____ T.

(Take $\epsilon_o = 8.85 \times 10^{-12} \mathrm{C^2/Nm^2}$ and $c = 3 \times 10^8\ \mathrm{m/s}$)

(A) 1.83

(B) 2.0

(C) 5.5

(D) 18.3

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Q14. JEE Main 2026 (Online) 21st January Morning Shift

Open solution

The electric field in a plane electromagnetic wave is given by :

$E_y = 69 \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right] \mathrm{V/m}.$

The expression for magnetic field associated with this electromagnetic wave is ____ T.

(A) $B_z = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

(B) $B_z = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

(C) $B_y = 2.3 \times 10^{-7} \sin\left[0.6 \times 10^3 x - 1.8 \times 10^{11} t\right]$

(D) $B_y = 69 \sin\left[0.6 \times 10^3 x + 1.8 \times 10^{11} t\right]$

Q15. JEE Main 2025 (Online) 7th April Evening Shift

Open solution

The unit of $\sqrt{\frac{2I}{\varepsilon_0 c}}$ is :

(I = intensity of an electromagnetic wave, c = speed of light)

(A) Vm (B) NC^{-1} (C) NC (D) Nm

Q16. JEE Main 2025 (Online) 7th April Morning Shift

Open solution

If ϵ_0 denotes the permittivity of free space and Φ_E is the flux of the electric field through the area bounded by the closed surface, then dimensions of $\left(\epsilon_0 \frac{d\phi_E}{dt}\right)$ are that of :

(A) electric charge (B) electric field

(C) electric current (D) electric potential

Q17. JEE Main 2025 (Online) 29th January Evening Shift

Open solution

A plane electromagnetic wave propagates along the + x direction in free space. The components of the electric field, $\vec{E}$ and magnetic field, $\vec{B}$ vectors associated with the wave in Cartesian frame are

(A) E_x, B_y (B) E_y, B_x (C) E_y, B_z (D) E_z, B_y

Q18. JEE Main 2025 (Online) 29th January Morning Shift

Open solution

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A) : Electromagnetic waves carry energy but not momentum.

Reason (R) : Mass of a photon is zero.

In the light of the above statements, choose the **most appropriate answer** from the options given below :

(A) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(B) Both **(A)** and **(R)** are true but **(R)** is not the correct explanation of **(A)**

(C) **(A)** is false but **(R)** is true

(D) **(A)** is true but **(R)** is false

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Q19. JEE Main 2025 (Online) 28th January Evening Shift

Open solution

The magnetic field of an E.M. wave is given by $\vec{B} = \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)30\sin\left[\omega\left(t - \frac{z}{c}\right)\right]$ (S.I. Units).

The corresponding electric field in S.I. units is:

(A) $\overrightarrow{\mathrm{E}} = \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)30\mathrm{c}\sin\left[\omega\left(\mathrm{t} + \frac{z}{\mathrm{c}}\right)\right]$

(B) $\overrightarrow{\mathrm{E}} = \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)30\mathrm{c}\sin\left[\omega\left(\mathrm{t} - \frac{z}{\mathrm{c}}\right)\right]$

(C) $\overrightarrow{\mathrm{E}} = \left(\frac{\sqrt{3}}{2}\hat{i} - \frac{1}{2}\hat{j}\right)30\mathrm{c}\sin\left[\omega\left(\mathrm{t} + \frac{z}{\mathrm{c}}\right)\right]$

(D) $\overrightarrow{\mathrm{E}} = \left(\frac{3}{4}\hat{i} + \frac{1}{4}\hat{j}\right)30\mathrm{c}\cos\left[\omega\left(\mathrm{t} - \frac{z}{\mathrm{c}}\right)\right]$

Q20. JEE Main 2025 (Online) 28th January Morning Shift

Open solution

Due to presence of an em-wave whose electric component is given by $E = 100\sin(\omega t - kx)\mathrm{NC}^{-1}$ a cylinder of length 200 cm holds certain amount of em-energy inside it. If another cylinder of same length but half diameter than previous one holds same amount of em-energy, the magnitude of the electric field of the corresponding em-wave should be modified as

(A) $50\sin(\omega\mathrm{t} - \mathrm{kx})\mathrm{NC}^{-1}$

(B) $400\sin(\omega\mathrm{t} - \mathrm{kx})\mathrm{NC}^{-1}$

(C) $200\sin(\omega t - kx)\mathrm{NC}^{-1}$

(D) $25\sin(\omega\mathrm{t} - \mathrm{kx})\mathrm{NC}^{-1}$

Q21. JEE Main 2025 (Online) 24th January Evening Shift

Open solution

Arrange the following in the ascending order of wavelength (λ) :

(A) Microwaves (λ_1)

(B) Ultraviolet rays (λ_2)

(C) Infrared rays (λ_3)

(D) X-rays (λ_4)

Choose the most appropriate answer from the options given below :

(A) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$

(B) $\lambda_4 < \lambda_2 < \lambda_3 < \lambda_1$

(C) $\lambda_3 < \lambda_4 < \lambda_2 < \lambda_1$

(D) $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

Q22. JEE Main 2025 (Online) 23rd January Evening Shift

Open solution

A plane electromagnetic wave of frequency 20 MHz travels in free space along the $+x$ direction. At a particular point in space and time, the electric field vector of the wave is $\mathrm{E}_y = 9.3\mathrm{Vm}^{-1}$. Then, the magnetic field vector of the wave at that point is

(A) $\mathrm{B}_z = 1.55 \times 10^{-8}\ \mathrm{T}$

(B) $\mathrm{B}_z = 6.2 \times 10^{-8}\ \mathrm{T}$

(C) $\mathrm{B}_z = 3.1 \times 10^{-8}\ \mathrm{T}$

(D) $\mathrm{B}_z = 9.3 \times 10^{-8}\ \mathrm{T}$

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Q23. JEE Main 2025 (Online) 23rd January Morning Shift

The electric field of an electromagnetic wave in free space is $\overrightarrow{\mathrm{E}} = 57\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](4\hat{i}-3\hat{j})N/C$. The associated magnetic field in Tesla is

(A) $\overrightarrow{\mathrm{B}} = \frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](\hat{k})$

(B) $\overrightarrow{\mathrm{B}} = \frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](5\hat{k})$

(C) $\overrightarrow{\mathrm{B}} = -\frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](\hat{k})$

(D) $\overrightarrow{\mathrm{B}} = -\frac{57}{3\times10^{8}}\cos\left[7.5\times10^{6}\mathrm{t} - 5\times10^{-3}(3x+4y)\right](5\hat{k})$

Open solution

Q24. JEE Main 2024 (Online) 9th April Evening Shift

The magnetic field in a plane electromagnetic wave is $\mathrm{B_y} = \left(3.5\times10^{-7}\right)\sin\left(1.5\times10^{3}x + 0.5\times10^{11}t\right)\mathrm{T}$. The corresponding electric field will be :

(A) $E_z = 105\sin\left(1.5\times10^{3}x + 0.5\times10^{11}t\right)\mathrm{Vm^{-1}}$

(B) $E_y = 10.5\sin\left(1.5\times10^{3}x + 0.5\times10^{11}t\right)\mathrm{Vm^{-1}}$

(C) $E_y = 1.17\sin\left(1.5\times10^{3}x + 0.5\times10^{11}t\right)\mathrm{Vm^{-1}}$

(D) $E_z = 1.17\sin\left(1.5\times10^{3}x + 0.5\times10^{11}t\right)\mathrm{Vm^{-1}}$

Open solution

Q25. JEE Main 2024 (Online) 9th April Morning Shift

A plane EM wave is propagating along x direction. It has a wavelength of $4\ \mathrm{mm}$. If electric field is in y direction with the maximum magnitude of $60\ \mathrm{Vm^{-1}}$, the equation for magnetic field is :

(A) $\mathrm{B}_z = 2\times10^{-7}\sin\left[\frac{\pi}{2}\left(x - 3\times10^{8}\mathrm{t}\right)\right]\hat{\mathrm{k}}\mathrm{T}$

(B) $\mathrm{B}_z = 2\times10^{-7}\sin\left[\frac{\pi}{2}\times10^{3}\left(x - 3\times10^{8}\mathrm{t}\right)\right]\hat{\mathrm{k}}\mathrm{T}$

(C) $\mathrm{B}_z = 60\sin\left[\frac{\pi}{2}\left(x - 3\times10^{8}\mathrm{t}\right)\right]\hat{\mathrm{k}}\mathrm{T}$

(D) $\mathrm{B}_x = 60\sin\left[\frac{\pi}{2}\left(x - 3\times10^{8}\mathrm{t}\right)\right]\hat{\mathrm{i}}\mathrm{T}$

Open solution

Q26. JEE Main 2024 (Online) 8th April Morning Shift

Average force exerted on a non-reflecting surface at normal incidence is $2.4\times10^{-4}\ \mathrm{N}$. If $360\ \mathrm{W/cm^2}$ is the light energy flux during span of 1 hour 30 minutes, then the area of the surface is:

(A) $20\ \mathrm{m^2}$ (B) $0.2\ \mathrm{m^2}$ (C) $0.1\ \mathrm{m^2}$ (D) $0.02\ \mathrm{m^2}$

Open solution

Q27. JEE Main 2024 (Online) 6th April Evening Shift

In the given electromagnetic wave $\mathrm{E_y} = 600\sin(\omega t - \mathrm{kx})\mathrm{Vm^{-1}}$, intensity of the associated light beam is (in $\mathrm{W/m^2}$) : (Given $\epsilon_0 = 9\times10^{-12}\mathrm{C^2\,N^{-1}\,m^{-2}}$)

(A) 486 (B) 729 (C) 243 (D) 972

Open solution

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Q28. JEE Main 2024 (Online) 6th April Morning Shift

Electromagnetic waves travel in a medium with speed of $1.5 \times 10^8\ \mathrm{m\,s^{-1}}$. The relative permeability of the medium is 2.0. The relative permittivity will be:

Open solution

(A) 4 (B) 1 (C) 2 (D) 5

Q29. JEE Main 2024 (Online) 5th April Evening Shift

Match List I with List II :

Open solution

	LIST I EM-Wave		LIST II Wavelength Range
A.	Infra-red	I.	$< 10^{-3}$ nm
B.	Ultraviolet	II.	400 nm to 1 nm
C.	X-rays	III.	1 mm to 700 nm
D.	Gamma rays	IV.	1 nm to 10^{-3} nm

Choose the correct answer from the options given below :

(A) (A)-(I), (B)-(III), (C)-(II), (D)-(IV)
(B) (A)-(III), (B)-(II), (C)-(IV), (D)-(I)
(C) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)
(D) (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

Q30. JEE Main 2024 (Online) 5th April Morning Shift

An alternating voltage of amplitude $40\ \mathrm{V}$ and frequency $4\ \mathrm{kHz}$ is applied directly across the capacitor of $12\mu\mathrm{F}$. The maximum displacement current between the plates of the capacitor is nearly :

Open solution

(A) 10 A (B) 8 A (C) 13 A (D) 12 A

Q31. JEE Main 2024 (Online) 4th April Evening Shift

Arrange the following in the ascending order of wavelength:

A. Gamma rays (λ_1)

B. x - rays (λ_2)

C. Infrared waves (λ_3)

D. Microwaves (λ_4)

Choose the most appropriate answer from the options given below

Open solution

(A) $\lambda_1 < \lambda_2 < \lambda_3 < \lambda_4$
(B) $\lambda_2 < \lambda_1 < \lambda_4 < \lambda_3$
(C) $\lambda_4 < \lambda_3 < \lambda_2 < \lambda_1$
(D) $\lambda_4 < \lambda_3 < \lambda_1 < \lambda_2$

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Q32. JEE Main 2024 (Online) 4th April Morning Shift

Open solution

The electric field in an electromagnetic wave is given by $\overrightarrow{\mathrm{E}}=\hat{i}40\cos\omega(\mathrm{t}-z/\mathrm{c})\mathrm{NC}^{-1}$. The magnetic field induction of this wave is (in SI unit) :

(A) $\overrightarrow{\mathrm{B}}=\hat{j}\frac{40}{\mathrm{c}}\cos\omega(\mathrm{t}-z/\mathrm{c})$

(B) $\overrightarrow{\mathrm{B}}=\hat{i}\frac{40}{\mathrm{c}}\cos\omega(\mathrm{t}-z/\mathrm{c})$

(C) $\vec{B}=\hat{j}40\cos\omega(t-z/c)$

(D) $\overrightarrow{\mathrm{B}}=\hat{k}\frac{40}{\mathrm{c}}\cos\omega(\mathrm{t}-z/\mathrm{c})$

Q33. JEE Main 2024 (Online) 1st February Evening Shift

Open solution

If frequency of electromagnetic wave is $60\ \mathrm{MHz}$ and it travels in air along z direction then the corresponding electric and magnetic field vectors will be mutually perpendicular to each other and the wavelength of the wave (in m) is :

(A) 2.5 (B) 5 (C) 10 (D) 2

Q34. JEE Main 2024 (Online) 1st February Morning Shift

Open solution

A parallel plate capacitor has a capacitance $\mathrm{C}=200\ \mathrm{pF}$. It is connected to $230\ \mathrm{V}$ ac supply with an angular frequency $300\ \mathrm{rad/s}$. The rms value of conduction current in the circuit and displacement current in the capacitor respectively are :

(A) $14.3\ \mu\mathrm{A}$ and $143\ \mu\mathrm{A}$

(B) $13.8\ \mu\mathrm{A}$ and $13.8\ \mu\mathrm{A}$

(C) $13.8\ \mu\mathrm{A}$ and $138\ \mu\mathrm{A}$

(D) $1.38\ \mu\mathrm{A}$ and $1.38\ \mu\mathrm{A}$

Q35. JEE Main 2024 (Online) 31st January Evening Shift

Open solution

Given below are two statements:

Statement I: Electromagnetic waves carry energy as they travel through space and this energy is equally shared by the electric and magnetic fields.

Statement II: When electromagnetic waves strike a surface, a pressure is exerted on the surface.

In the light of the above statements, choose the most appropriate answer from the options given below:

(A) Statement I is incorrect but Statement II is correct.

(B) Both Statement I and Statement II are correct.

(C) Statement I is correct but Statement II is incorrect.

(D) Both Statement I and Statement II are incorrect.

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Q36. JEE Main 2024 (Online) 31st January Morning Shift

Open solution

In a plane EM wave, the electric field oscillates sinusoidally at a frequency of 5×10^{10} Hz and an amplitude of $50\ \mathrm{Vm}^{-1}$. The total average energy density of the electromagnetic field of the wave is : [Use $\varepsilon_0 = 8.85 \times 10^{-12}\mathrm{C}^2/\mathrm{Nm}^2$]

(A) $4.425 \times 10^{-8}\mathrm{Jm}^{-3}$

(B) $2.212 \times 10^{-10}\mathrm{Jm}^{-3}$

(C) $2.212 \times 10^{-8}\mathrm{Jm}^{-3}$

(D) $1.106 \times 10^{-8}\mathrm{Jm}^{-3}$

Q37. JEE Main 2024 (Online) 30th January Evening Shift

Open solution

Match List I with List II

	List I		List II
(A)	Gauss's law of magnetostatics	(I)	$\oint \vec{E} \cdot \vec{da} = \frac{1}{\varepsilon_0} \int \rho dV$
(B)	Faraday's law of electromagnetic induction	(II)	$\oint \vec{B} \cdot \vec{da} = 0$
(C)	Ampere's law	(III)	$\int \vec{E} \cdot \vec{dl} = \frac{-d}{dt} \int \vec{B} \cdot \vec{da}$
(D)	Gauss's law of electrostatics	(IV)	$\oint \vec{B} \cdot \vec{dl} = \mu_0 I$

Choose the correct answer from the options given below:

(A) A-III, B-IV, C-I, D-II

(B) A-IV, B-II, C-III, D-I

(C) A-II, B-III, C-IV, D-I

(D) A-I, B-III, C-IV, D-II

Q38. JEE Main 2024 (Online) 30th January Morning Shift

Open solution

The electric field of an electromagnetic wave in free space is represented as $\overrightarrow{\mathrm{E}} = \mathrm{E}_0 \cos(\omega \mathrm{t} - \mathrm{kz})\hat{i}$. The corresponding magnetic induction vector will be :

(A) $\overrightarrow{\mathrm{B}} = \mathrm{E}_0 \mathrm{C} \cos(\omega \mathrm{t} + \mathrm{k}z)\hat{j}$

(B) $\overrightarrow{\mathrm{B}} = \frac{\mathrm{E}_0}{\mathrm{C}} \cos(\omega \mathrm{t} - \mathrm{kz})\hat{j}$

(C) $\overrightarrow{\mathrm{B}} = \mathrm{E}_0 \mathrm{C} \cos(\omega \mathrm{t} - \mathrm{k}z)\hat{j}$

(D) $\overrightarrow{\mathrm{B}} = \frac{\mathrm{E}_0}{\mathrm{C}} \cos(\omega \mathrm{t} + \mathrm{kz})\hat{j}$

Q39. JEE Main 2024 (Online) 29th January Evening Shift

Open solution

A plane electromagnetic wave of frequency $35\ \mathrm{MHz}$ travels in free space along the X-direction. At a particular point (in space and time) $\vec{E} = 9.6\hat{j}\ \mathrm{V/m}$. The value of magnetic field at this point is :

(A) $9.6\hat{j}T$

(B) $3.2 \times 10^{-8}\hat{i}T$

(C) $9.6 \times 10^{-8}\hat{k}T$

(D) $3.2 \times 10^{-8}\hat{k}T$

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Q40. JEE Main 2024 (Online) 29th January Morning Shift

Open solution

Match List I with List II

	List - I		List - II
(A)	$\oint \vec{B} \cdot \overrightarrow{dl} = \mu_o i_c + \mu_o \varepsilon_o \frac{d\phi_E}{dt}$	(I)	Gauss' law for electricity
(B)	$\oint \vec{E} \cdot \overrightarrow{dl} = \frac{d\phi_B}{dt}$	(II)	Gauss' law for magnetism
(C)	$\oint \vec{E} \cdot \overrightarrow{dA} = \frac{Q}{\varepsilon_o}$	(III)	Faraday law
(D)	$\oint \vec{B} \cdot \overrightarrow{dA} = 0$	(IV)	Ampere - Maxwell law

Choose the correct answer from the options given below:

(A) A-IV, B-III, C-I, D-II

(B) A-I, B-II, C-III, D-IV

(C) A-IV, B-I, C-III, D-II

(D) A-II, B-III, C-I, D-IV

Q41. JEE Main 2024 (Online) 27th January Evening Shift

Open solution

An object is placed in a medium of refractive index 3 . An electromagnetic wave of intensity $6 \times 10^8\ \mathrm{W/m^2}$ falls normally on the object and it is absorbed completely. The radiation pressure on the object would be (speed of light in free space $= 3 \times 10^8\ \mathrm{m/s}$) :

(A) $6\ \mathrm{Nm^{-2}}$ **(B)** $36\ \mathrm{Nm^{-2}}$ **(C)** $18\ \mathrm{Nm^{-2}}$ **(D)** $2\ \mathrm{Nm^{-2}}$

Q42. JEE Main 2024 (Online) 27th January Morning Shift

Open solution

A plane electromagnetic wave propagating in x-direction is described by

$E_y = \left(200 \mathrm{Vm^{-1}}\right) \sin\left[1.5 \times 10^7 t - 0.05x\right]$;

The intensity of the wave is :

(Use $\epsilon_0 = 8.85 \times 10^{-12} \mathrm{C^2\ N^{-1}\ m^{-2}}$)

(A) $35.4\ \mathrm{Wm^{-2}}$ **(B)** $53.1\ \mathrm{Wm^{-2}}$ **(C)** $26.6\ \mathrm{Wm^{-2}}$ **(D)** $106.2\ \mathrm{Wm^{-2}}$

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Q43. JEE Main 2023 (Online) 15th April Morning Shift

Open solution

Match **List I** with **List II** of Electromagnetic waves with corresponding wavelength range :

List I	List II
(A) Microwave	(I) $400\ \text{nm}$ to $1\ \text{nm}$
(B) Ultraviolet	(II) $1\ \text{nm}$ to $10^{-3}\ \text{nm}$
(C) X-Ray	(III) $1\ \text{mm}$ to $700\ \text{nm}$
(D) Infra-red	(IV) $0.1\ \text{m}$ to $1\ \text{mm}$

Choose the correct answer from the options given below:

(A) (A)-(I), (B)-(IV), (C)-(II), (D)-(III)

(B) (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

(C) (A)-(IV), (B)-(I), (C)-(III), (D) -(II)

(D) (A)-(IV), (B)-(II), (C)-(I), (D)-(III)

Q44. JEE Main 2023 (Online) 13th April Evening Shift

Open solution

In an electromagnetic wave, at an instant and at particular position, the electric field is along the negative z-axis and magnetic field is along the positive x-axis. Then the direction of propagation of electromagnetic wave is:

(A) at 45° angle from positive y-axis

(B) positive y-axis

(C) negative y-axis

(D) positive z-axis

Q45. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

The source of time varying magnetic field may be

(A) a permanent magnet

(B) an electric field changing linearly with time

(C) direct current

(D) a decelerating charge particle

(E) an antenna fed with a digital signal

Choose the correct answer from the options given below:

(A) (D) only

(B) (A) only

(C) (B) and (D) only

(D) (C) and (E) only

Q46. JEE Main 2023 (Online) 13th April Morning Shift

Open solution

Which of the following Maxwell's equation is valid for time varying conditions but not valid for static conditions :

(A) $\oint \overrightarrow{\mathrm{E}} \cdot \overrightarrow{dl} = 0$

(B) $\oint \vec{B} \cdot \overrightarrow{dl} = \mu_0 I$

(C) $\oint \vec{E} \cdot \overrightarrow{dl} = -\frac{\partial \phi_B}{\partial t}$

(D) $\oint \vec{D} \cdot \overrightarrow{dA} = Q$

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Q47. JEE Main 2023 (Online) 12th April Morning Shift

Open solution

Given below are two statements: one is labelled as Assertion $\mathbf{A}$ and the other is labelled as Reason $\mathbf{R}$

Assertion A : EM waves used for optical communication have longer wavelengths than that of microwave, employed in Radar technology.

Reason R : Infrared EM waves are more energetic than microwaves, (used in Radar)

In the light of given statements, choose the correct answer from the options given below.

(A) Both A and R are true but R is NOT the correct explanation of A

(B) A is true but R is false

(C) Both A and R are true and R is the correct explanation of A

(D) A is false but R is true

Q48. JEE Main 2023 (Online) 11th April Evening Shift

Open solution

A plane electromagnetic wave of frequency $20\ \mathrm{MHz}$ propagates in free space along x-direction. At a particular space and time, $\overrightarrow{\mathrm{E}} = 6.6\hat{j}\ \mathrm{V/m}$. What is $\overrightarrow{\mathrm{B}}$ at this point?

(A) $-2.2 \times 10^{-8}\hat{i}T$ (B) $2.2 \times 10^{-8}\hat{i}T$ (C) $2.2 \times 10^{-8}\hat{k}T$ (D) $-2.2 \times 10^{-8}\hat{k}T$

Q49. JEE Main 2023 (Online) 11th April Morning Shift

Open solution

The electric field in an electromagnetic wave is given as

$\overrightarrow{\mathrm{E}} = 20\sin\omega\left(\mathrm{t} - \frac{x}{\mathrm{c}}\right)\overrightarrow{\mathrm{j}}\mathrm{NC}^{-1}$

where ω and c are angular frequency and velocity of electromagnetic wave respectively. The energy contained in a volume of $5 \times 10^{-4}\ \mathrm{m}^3$ will be

(Given $\varepsilon_0 = 8.85 \times 10^{-12}\mathrm{C}^2/\mathrm{Nm}^2$)

(A) $17.7 \times 10^{-13}\ \mathrm{J}$ (B) $28.5 \times 10^{-13}\ \mathrm{J}$ (C) $8.85 \times 10^{-13}\ \mathrm{J}$ (D) $88.5 \times 10^{-13}\ \mathrm{J}$

Q50. JEE Main 2023 (Online) 10th April Evening Shift

Open solution

The amplitude of magnetic field in an electromagnetic wave propagating along y-axis is $6.0 \times 10^{-7}\ \mathrm{T}$. The maximum value of electric field in the electromagnetic wave is

(A) $6.0 \times 10^{-7}\ \mathrm{Vm}^{-1}$ (B) $5 \times 10^{14}\ \mathrm{Vm}^{-1}$

(C) $180\ \mathrm{Vm}^{-1}$ (D) $2 \times 10^{15}\ \mathrm{Vm}^{-1}$

Q51. JEE Main 2023 (Online) 10th April Morning Shift

Open solution

The energy of an electromagnetic wave contained in a small volume oscillates with

(A) double the frequency of the wave (B) the frequency of the wave

(C) half the frequency of the wave (D) zero frequency

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Q52. JEE Main 2023 (Online) 8th April Evening Shift

The waves emitted when a metal target is bombarded with high energy electrons are

(A) Infrared rays (B) Radio Waves (C) Microwaves (D) X-rays

Open solution

Q53. JEE Main 2023 (Online) 6th April Evening Shift

The energy density associated with electric field $\vec{E}$ and magnetic field $\vec{B}$ of an electromagnetic wave in free space is given by (ϵ_0- permittivity of free space, μ_0- permeability of free space)

(A) $U_E = \frac{\epsilon_0 E^2}{2}, U_B = \frac{B^2}{2\mu_0}$ (B) $U_E = \frac{E^2}{2\epsilon_0}, U_B = \frac{\mu_0 B^2}{2}$

(C) $U_E = \frac{\epsilon_0 E^2}{2}, U_B = \frac{\mu_0 B^2}{2}$ (D) $U_E = \frac{E^2}{2\epsilon_0}, U_B = \frac{B^2}{2\mu_0}$

Open solution

Q54. JEE Main 2023 (Online) 6th April Morning Shift

For the plane electromagnetic wave given by $E = E_0 \sin(\omega t - kx)$ and $B = B_0 \sin(\omega t - kx)$, the ratio of average electric energy density to average magnetic energy density is

(A) 1 (B) 4 (C) 2 (D) 1/2

Open solution

Q55. JEE Main 2023 (Online) 1st February Evening Shift

The ratio of average electric energy density and total average energy density of electromagnetic wave is :

(A) 1 (B) 3 (C) 2 (D) $\frac{1}{2}$

Open solution

Q56. JEE Main 2023 (Online) 1st February Morning Shift

Match List I with List II :

	List I		List II
A.	Microwaves	I.	Radioactive decay of the nucleus
B.	Gamma rays	II.	Rapid acceleration and deceleration of electron in aerials
C.	Radio waves	III.	Inner shell electrons
D.	X-rays	IV.	Klystron valve

Choose the correct answer from the options given below :

(A) A-I, B-III, C-IV, D-II (B) A-IV, B-III, C-II, D-I

(C) A-IV, B-I, C-II, D-III (D) A-I, B-II, C-III, D-IV

Open solution

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Q57. JEE Main 2023 (Online) 31st January Evening Shift

Open solution

Match List **I** with List **II**

LIST I		LIST II	
A.	Microwaves	I.	Physiotherapy
B.	UV rays	II.	Treatment of cancer
C.	Infra-red light	III.	Lasik eye surgery
D.	X-ray	IV.	Aircraft navigation

Choose the correct answer from the options given below:

(A) A - IV, B - III, C - I, D - II

(B) A - II, B - IV, C - III, D - I

(C) A - III, B - II, C - I, D - IV

(D) A - IV, B - I, C - II, D - III

Q58. JEE Main 2023 (Online) 30th January Evening Shift

Open solution

A point source of $100\ \mathrm{W}$ emits light with 5% efficiency. At a distance of $5\ \mathrm{m}$ from the source, the intensity produced by the electric field component is:

(A) $\frac{1}{40\pi}\frac{W}{m^2}$ (B) $\frac{1}{10\pi}\frac{W}{m^2}$ (C) $\frac{1}{20\pi}\frac{W}{m^2}$ (D) $\frac{1}{2\pi}\frac{W}{m^2}$

Q59. JEE Main 2023 (Online) 29th January Evening Shift

Open solution

Given below are two statements :

Statement I : Electromagnetic waves are not deflected by electric and magnetic field.

Statement II : The amplitude of electric field and the magnetic field in electromagnetic waves are related to each other as $E_0 = \sqrt{\frac{\mu_0}{\varepsilon_0}}B_0$.

In the light of the above statements, choose the correct answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true and Statement II is false

(C) Both Statement I and Statement II are false

(D) Statement I is false but Statement II is true

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Q60. JEE Main 2023 (Online) 29th January Morning Shift

Open solution

Which of the following are true?

A. Speed of light in vacuum is dependent on the direction of propagation.

B. Speed of light in a medium is independent of the wavelength of light.

C. The speed of light is independent of the motion of the source.

D. The speed of light in a medium is independent of intensity.

Choose the correct answer from the options given below:

(A) B and D only (B) B and C only (C) A and C only (D) C and D only

Q61. JEE Main 2023 (Online) 25th January Evening Shift

Open solution

Match List I with List II

	List I		List II
A.	Gauss's Law in Electrostatics	I.	$\oint \vec{E} . \vec{dl} = -\frac{d\phi_B}{dt}$
B.	Faraday's Law	II.	$\oint \vec{B} . \vec{dA} = 0$
C.	Gauss's Law in Magnetism	III.	$\oint \vec{B} . \vec{dl} = \mu_0 i_c + \mu_0 \in_0 \frac{d\phi_E}{dt}$
D.	Ampere-Maxwell Law	IV.	$\oint \vec{E} . \vec{ds} = \frac{q}{\in_0}$

Choose the correct answer from the options given below :

(A) A-I, B-II, C-III, D-IV (B) A-III, B-IV, C-I, D-II

(C) A-IV, B-I, C-II, D-III (D) A-II, B-III, C-IV, D-I

Q62. JEE Main 2023 (Online) 25th January Morning Shift

Open solution

An electromagnetic wave is transporting energy in the negative z direction. At a certain point and certain time the direction of electric field of the wave is along positive y direction. What will be the direction of the magnetic field of the wave at that point and instant?

(A) Negative direction of y (B) Positive direction of z

(C) Positive direction of x (D) Negative direction x

Q63. JEE Main 2023 (Online) 24th January Evening Shift

Open solution

The electric field and magnetic field components of an electromagnetic wave going through vacuum is described by

$\mathrm{E_x = E_o \sin(kz - \omega t)}$

$\mathrm{B_y = B_o \sin(kz - \omega t)}$

Then the correct relation between E_0 and B_0 is given by

(A) $\mathrm{E_o B_o = \omega k}$ (B) $\mathrm{E_0 = kB_0}$ (C) $\mathrm{kE_0 = \omega B_0}$ (D) $\mathrm{\omega E_0 = kB_0}$

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Q64. JEE Main 2023 (Online) 24th January Morning Shift

If $\overrightarrow{E}$ and $\overrightarrow{K}$ represent electric field and propagation vectors of the EM waves in vacuum, then magnetic field vector is given by :

(ω - angular frequency) :

(A) $\frac{1}{\omega}\left(\overrightarrow{K} \times \overrightarrow{E}\right)$

(B) $\overrightarrow{K} \times \overrightarrow{E}$

(C) $\omega\left(\overrightarrow{K} \times \overrightarrow{E}\right)$

(D) $\omega\left(\overrightarrow{E} \times \overrightarrow{K}\right)$

Open solution

Q65. JEE Main 2022 (Online) 30th July Evening Shift

A plane electromagnetic wave propagating in free space has amplitude of electric field of $900\ \frac{\mathrm{N}}{\mathrm{C}}$. The amplitude of magnetic field will be :_____

(A) $9\ \mu\mathrm{T}$

(B) $27\ \mu\mathrm{T}$

(C) $3\ \mu\mathrm{T}$

(D) $270\ \mathrm{GT}$

Open solution

Q66. JEE Main 2022 (Online) 29th July Morning Shift

Match List - I with List - II :

	List - I		List - II
(a)	UV rays	(i)	Diagnostic tool in medicine
(b)	X-rays	(ii)	Water purification
(c)	Microwave	(iii)	Communication, Radar
(d)	Infrared wave	(iv)	Improving visibility in foggy days

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(C) (a)-(ii), (b)-(iv), (c)-(iii), (d)-(i)

(D) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

Open solution

Q67. JEE Main 2022 (Online) 28th July Evening Shift

Sun light falls normally on a surface of area $36\ \mathrm{cm}^2$ and exerts an average force of $7.2 \times 10^{-9}\ \mathrm{N}$ within a time period of 20 minutes. Considering a case of complete absorption, the energy flux of incident light is

(A) $25.92 \times 10^2\ \mathrm{W/cm}^2$

(B) $8.64 \times 10^{-6}\ \mathrm{W/cm}^2$

(C) $6.0\ \mathrm{W/cm}^2$

(D) $0.06\ \mathrm{W/cm}^2$

Open solution

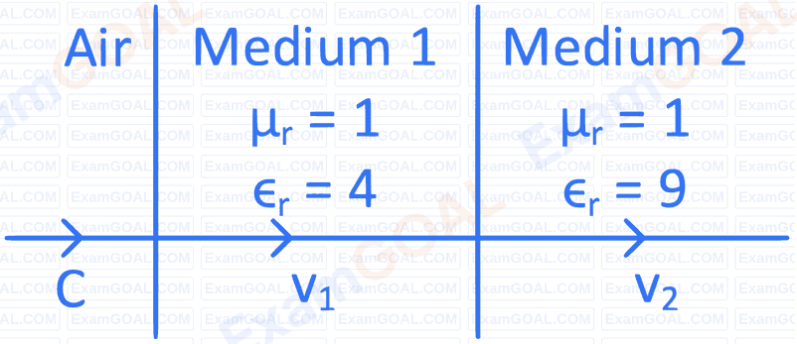
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Q68. JEE Main 2022 (Online) 28th July Morning Shift

Open solution

As shown in the figure, after passing through the medium 1 . The speed of light v_2 in medium 2 will be :

(Given $c = 3 \times 10^8\ ms^{-1}$)



(A) $1.0 \times 10^8\ ms^{-1}$

(B) $0.5 \times 10^8\ ms^{-1}$

(C) $1.5 \times 10^8\ ms^{-1}$

(D) $3.0 \times 10^8\ ms^{-1}$

Q69. JEE Main 2022 (Online) 27th July Evening Shift

Open solution

Identify the correct statements from the following descriptions of various properties of electromagnetic waves.

(A) In a plane electromagnetic wave electric field and magnetic field must be perpendicular to each other and direction of propagation of wave should be along electric field or magnetic field.

(B) The energy in electromagnetic wave is divided equally between electric and magnetic fields.

(C) Both electric field and magnetic field are parallel to each other and perpendicular to the direction of propagation of wave.

(D) The electric field, magnetic field and direction of propagation of wave must be perpendicular to each other.

(E) The ratio of amplitude of magnetic field to the amplitude of electric field is equal to speed of light.

Choose the most appropriate answer from the options given below :

(A) (D) only

(B) (B) and (D) only

(C) (B), (C) and (E) only

(D) (A), (B) and (E) only

Q70. JEE Main 2022 (Online) 27th July Morning Shift

Open solution

A beam of light travelling along X-axis is described by the electric field $E_y = 900 \sin \omega(t - x/c)$. The ratio of electric force to magnetic force on a charge q moving along Y-axis with a speed of $3 \times 10^7\ ms^{-1}$ will be :

(Given speed of light $= 3 \times 10^8\ ms^{-1}$)

(A) 1 : 1

(B) 1 : 10

(C) 10 : 1

(D) 1 : 2

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Q71. JEE Main 2022 (Online) 26th July Evening Shift

Open solution

The oscillating magnetic field in a plane electromagnetic wave is given by

$B_y = 5 \times 10^{-6} \sin 1000\pi \left(5x - 4 \times 10^8 t\right) T$. The amplitude of electric field will be :

(A) $15 \times 10^2\ \mathrm{Vm}^{-1}$ (B) $5 \times 10^{-6}\ \mathrm{Vm}^{-1}$

(C) $16 \times 10^{12}\ \mathrm{Vm}^{-1}$ (D) $4 \times 10^2\ \mathrm{Vm}^{-1}$

Q72. JEE Main 2022 (Online) 26th July Morning Shift

Open solution

The magnetic field of a plane electromagnetic wave is given by :

$\overrightarrow{\mathrm{B}} = 2 \times 10^{-8} \sin\left(0.5 \times 10^3 x + 1.5 \times 10^{11} \mathrm{t}\right) \hat{j}\ \mathrm{T}$.

The amplitude of the electric field would be :

(A) $6\ \mathrm{Vm}^{-1}$ along x-axis (B) $3\ \mathrm{Vm}^{-1}$ along z-axis

(C) $6\ \mathrm{Vm}^{-1}$ along z-axis (D) $2 \times 10^{-8}\ \mathrm{Vm}^{-1}$ along z-axis

Q73. JEE Main 2022 (Online) 25th July Evening Shift

Open solution

Light wave travelling in air along x-direction is given by $E_y = 540 \sin \pi \times 10^4 (x - ct)\ Vm^{-1}$. Then, the peak value of magnetic field of wave will be (Given c = 3×10^8 ms^{-1})

(A) 18×10^{-7} T (B) 54×10^{-7} T (C) 54×10^{-8} T (D) 18×10^{-8} T

Q74. JEE Main 2022 (Online) 30th June Morning Shift

Open solution

An expression for oscillating electric field in a plane electromagnetic wave is given as E_z = 300 sin($5\pi \times 10^3$x $- 3\pi \times 10^{11}$t) Vm^{-1}

Then, the value of magnetic field amplitude will be :

(Given : speed of light in Vacuum c = 3×10^8 ms^{-1})

(A) 1×10^{-6} T (B) 5×10^{-6} T (C) 18×10^9 T (D) 21×10^9 T

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Q75. JEE Main 2022 (Online) 28th June Evening Shift

Open solution

An EM wave propagating in x-direction has a wavelength of 8 mm. The electric field vibrating y-direction has maximum magnitude of 60 Vm^{-1}. Choose the correct equations for electric and magnetic fields if the EM wave is propagating in vacuum :

(A) $E_y = 60\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 2\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(B) $E_y = 60\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 2 \times 10^{-7}\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(C) $E_y = 2 \times 10^{-7}\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4} \times 10^3(x - 3 \times 10^8 t)\right]\hat{k}\ T$

(D) $E_y = 2 \times 10^{-7}\sin\left[\frac{\pi}{4} \times 10^4(x - 4 \times 10^8 t)\right]\hat{j}\ Vm^{-1}$

$B_z = 60\sin\left[\frac{\pi}{4} \times 10^4(x - 4 \times 10^8 t)\right]\hat{k}\ T$

Q76. JEE Main 2022 (Online) 28th June Morning Shift

Open solution

A radar sends an electromagnetic signal of electric field (E_0) = 2.25 V/m and magnetic field (B_0) = 1.5×10^{-8} T which strikes a target on line of sight at a distance of 3 km in a medium. After that, a part of signal (echo) reflects back towards the radar with same velocity and by same path. If the signal was transmitted at time t = 0 from radar, then after how much time echo will reach to the radar?

(A) 2.0×10^{-5} s (B) 4.0×10^{-5} s (C) 1.0×10^{-5} s (D) 8.0×10^{-5} s

Q77. JEE Main 2022 (Online) 27th June Evening Shift

Open solution

Given below are two statements :

Statement I : A time varying electric field is a source of changing magnetic field and vice-versa. Thus a disturbance in electric or magnetic field creates EM waves.

Statement II : In a material medium, the EM wave travels with speed $v = \frac{1}{\sqrt{\mu_0 \in_0}}$. In the light of the above statements, choose the correct answer from the options given below.

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is correct but Statement II is false

(D) Statement I is incorrect but Statement II is true

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Q78. JEE Main 2022 (Online) 27th June Morning Shift

Open solution

Match List-I with List-II :

	List - I		List - II
(a)	Ultraviolet rays	(i)	Study crystal structure
(b)	Microwaves	(ii)	Greenhouse effect
(c)	Infrared rays	(iii)	Sterilizing surgical instrument
(d)	X-rays	(iv)	Radar system

Choose the correct answer from the options given below :

(A) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

(B) (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)

(C) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(D) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

Q79. JEE Main 2022 (Online) 26th June Evening Shift

Open solution

Which is the correct ascending order of wavelengths?

(A) $\lambda_{\text{visible}} < \lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{microwave}}$

(B) $\lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(C) $\lambda_{\text{X-ray}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{visible}} < \lambda_{\text{microwave}}$

(D) $\lambda_{\text{microwave}} < \lambda_{\text{visible}} < \lambda_{\text{gamma-ray}} < \lambda_{\text{X-ray}}$

Q80. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

If Electric field intensity of a uniform plane electromagnetic wave is given as $E = -301.6\sin(kz - \omega t)\hat{a}_x + 452.4\sin(kz - \omega t)\hat{a}_y \frac{V}{m}$. Then magnetic intensity 'H' of this wave in Am $^{-1}$ will be :

[Given : Speed of light in vacuum $c = 3 \times 10^8$ ms^{-1}, Permeability of vacuum $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}]

(A) $+0.8\sin(kz - \omega t)\hat{a}_y + 0.8\sin(kz - \omega t)\hat{a}_x$

(B) $+1.0 \times 10^{-6}\sin(kz - \omega t)\hat{a}_y + 1.5 \times 10^{-6}(kz - \omega t)\hat{a}_x$

(C) $-0.8\sin(kz - \omega t)\hat{a}_y - 1.2\sin(kz - \omega t)\hat{a}_x$

(D) $-1.0 \times 10^{-6}\sin(kz - \omega t)\hat{a}_y - 1.5 \times 10^{-6}\sin(kz - \omega t)\hat{a}_x$

Q81. JEE Main 2022 (Online) 26th June Morning Shift

Open solution

In free space, an electromagnetic wave of 3 GHz frequency strikes over the edge of an object of size $\frac{\lambda}{100}$, where λ is the wavelength of the wave in free space. The phenomenon, which happens there will be :

(A) Reflection (B) Refraction (C) Diffraction (D) Scattering

Q82. JEE Main 2022 (Online) 25th June Evening Shift

Open solution

The electromagnetic waves travel in a medium at a speed of 2.0×10^8 m/s. The relative permeability of the medium is 1.0. The relative permittivity of the medium will be :

(A) 2.25 (B) 4.25 (C) 6.25 (D) 8.25

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Q83. JEE Main 2022 (Online) 25th June Morning Shift

The electric field in an electromagnetic wave is given by E = 56.5 sin ω(t $-$ x/c) NC^{-1}. Find the intensity of the wave if it is propagating along x-axis in the free space.

(Given : $\varepsilon_0 = 8.85 \times 10^{-12} C^2 N^{-1} m^{-2}$)

(A) 5.65 Wm^{-2}

(B) 4.24 Wm^{-2}

(C) 1.9×10^{-7} Wm^{-2}

(D) 56.5 Wm^{-2}

Open solution

Q84. JEE Main 2022 (Online) 24th June Evening Shift

An electric bulb is rated as 200 W. What will be the peak magnetic field at 4 m distance produced by the radiations coming from this bulb? Consider this bulb as a point source with 3.5% efficiency.

(A) 1.19×10^{-8}T (B) 1.71×10^{-8}T (C) 0.84×10^{-8}T (D) 3.36×10^{-8}T

Open solution

Q85. JEE Main 2022 (Online) 24th June Morning Shift

A plane electromagnetic wave travels in a medium of relative permeability 1.61 and relative permittivity 6.44. If magnitude of magnetic intensity is 4.5×10^{-2} Am^{-1} at a point, what will be the approximate magnitude of electric field intensity at that point?

(Given : Permeability of free space $\mu_0 = 4\pi \times 10^{-7}$ NA^{-2}, speed of light in vacuum c = 3×10^8 ms^{-1})

(A) 16.96 Vm^{-1}

(B) 2.25×10^{-2} Vm^{-1}

(C) 8.48 Vm^{-1}

(D) 6.75×10^6 Vm^{-1}

Open solution

Q86. JEE Main 2021 (Online) 1st September Evening Shift

Electric field of plane electromagnetic wave propagating through a non-magnetic medium is given by

E = 20cos(2×10^{10} t $-$ 200x) V/m. The dielectric constant of the medium is equal to : (Take μ_r = 1)

(A) 9 (B) 2 (C) $\frac{1}{3}$ (D) 3

Open solution

Q87. JEE Main 2021 (Online) 31st August Evening Shift

The magnetic field vector of an electromagnetic wave is given by $B = B_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz - \omega t)$; where $\hat{i}, \hat{j}$ represents unit vector along x and y-axis respectively. At t = 0s, two electric charges q_1 of 4π coulomb and q_2 of 2π coulomb located at $\left(0, 0, \frac{\pi}{k}\right)$ and $\left(0, 0, \frac{3\pi}{k}\right)$, respectively, have the same velocity of 0.5 c $\hat{i}$, (where c is the velocity of light). The ratio of the force acting on charge q_1 to q_2 is :-

(A) $2\sqrt{2} : 1$ (B) $1 : \sqrt{2}$ (C) 2 : 1 (D) $\sqrt{2} : 1$

Open solution

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Q88. JEE Main 2021 (Online) 27th August Morning Shift

Open solution

Electric field in a plane electromagnetic wave is given by E = 50 sin(500x $-$ 10 $\times$ 10^{10} t) V/m The velocity of electromagnetic wave in this medium is :

(Given C = speed of light in vacuum)

(A) $\frac{3}{2}$C (B) C (C) $\frac{2}{3}$C (D) $\frac{C}{2}$

Q89. JEE Main 2021 (Online) 26th August Evening Shift

Open solution

A light beam is described by $E = 800 \sin\omega\left(t - \frac{x}{c}\right)$. An electron is allowed to move normal to the propagation of light beam with a speed of 3 $\times$ 10^7 ms^{-1}. What is the maximum magnetic force exerted on the electron?

(A) 1.28 $\times$ 10^{-18} N (B) 1.28 $\times$ 10^{-21} N (C) 12.8 $\times$ 10^{-17} N (D) 12.8 $\times$ 10^{-18} N

Q90. JEE Main 2021 (Online) 27th July Morning Shift

Open solution

The relative permittivity of distilled water is 81. The velocity of light in it will be :

(Given μ_r = 1)

(A) 4.33 $\times$ 10^7 m/s (B) 2.33 $\times$ 10^7 m/s (C) 3.33 $\times$ 10^7 m/s (D) 5.33 $\times$ 10^7 m/s

Q91. JEE Main 2021 (Online) 25th July Morning Shift

Open solution

A linearly polarized electromagnetic wave in vacuum

$$E = 3.1\cos\left[(1.8)z - (5.4\times10^6)t\right]\hat{i}N/C$$

is incident normally on a perfectly reflecting wall at z = a. Choose the correct option

(A) The wavelength is 5.4 m

(B) The frequency of electromagnetic wave is 54 $\times$ 10^4 Hz.

(C) The transmitted wave will be $3.1\cos\left[(1.8)z - (5.4\times10^6)t\right]\hat{i}N/C$

(D) The reflected wave will be $3.1\cos\left[(1.8)z + (5.4\times10^6)t\right]\hat{i}N/C$

Q92. JEE Main 2021 (Online) 22th July Evening Shift

Open solution

Intensity of sunlight is observed as 0.092 Wm^{-2} at a point in free space. What will be the peak value of magnetic field at the point?

$(\varepsilon_0 = 8.85\times10^{-12}C^2N^{-1}m^{-2})$

(A) 2.77 $\times$ 10^{-8} T (B) 1.96 $\times$ 10^{-8} T (C) 8.31 T (D) 5.88 T

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Q93. JEE Main 2021 (Online) 20th July Evening Shift

In an electromagnetic wave the electric field vector and magnetic field vector are given as $\vec{E} = E_0\hat{i}$ and $\vec{B} = B_0\hat{k}$ respectively. The direction of propagation of electromagnetic wave is along :

(A) $\left(\hat{k}\right)$ (B) $\hat{j}$ (C) $\left(-\hat{k}\right)$ (D) $\left(-\hat{j}\right)$

Open solution

Q94. JEE Main 2021 (Online) 20th July Morning Shift

AC voltage V(t) = 20 sinωt of frequency 50 Hz is applied to a parallel plate capacitor. The separation between the plates is 2 mm and the area is 1 m^2. The amplitude of the oscillating displacement current for the applied AC voltage is _____. [Take ε_0 = 8.85 $\times$ 10^{-12} F/m]

(A) 55.58 μA (B) 21.14 μA (C) 27.79 μA (D) 83.37 μA

Open solution

Q95. JEE Main 2021 (Online) 18th March Evening Shift

A plane electromagnetic wave propagating along y-direction can have the following pair of electric field $\left(\vec{E}\right)$ and magnetic field $\left(\vec{B}\right)$ components.

(A) E_x, B_z or E_z, B_x (B) E_x, B_y or E_y, B_x (C) E_y, B_y or E_z, B_z (D) E_y, B_x or E_x, B_y

Open solution

Q96. JEE Main 2021 (Online) 18th March Morning Shift

A plane electromagnetic wave of frequency 100 MHz is travelling in vacuum along the x-direction. At a particular point in space and time, $\vec{B} = 2.0 \times 10^{-8}\hat{k}T$. (where, $\hat{k}$ is unit vector along z-direction) What is $\vec{E}$ at this point?

(A) 0.6 $\hat{j}$ V/m (B) 6.0 $\hat{k}$ V/m (C) 6.0 $\hat{j}$ V/m (D) 0.6 $\hat{k}$ V/m

Open solution

Q97. JEE Main 2021 (Online) 16th March Morning Shift

A plane electromagnetic wave of frequency 500 MHz is travelling in vacuum along y-direction. At a particular point in space and time,
$\vec{B}$ = 8.0 $\times$ 10^{-8} $\hat{z}$T. The value of electric field at this point is :

(speed of light = 3 $\times$ 10^8 ms^{-1})

$\hat{x}$, $\hat{y}$, $\hat{z}$ are unit vectors along x, y and z directions.

(A) 2.6 $\hat{x}$ V/m (B) -24 $\hat{x}$ V/m (C) 24 $\hat{x}$ V/m (D) -2.6 $\hat{y}$ V/m

Open solution

Q98. JEE Main 2021 (Online) 16th March Morning Shift

For an electromagnetic wave travelling in free space, the relation between average energy densities due to electric (U_e) and magnetic (U_m) fields is :

(A) $U_e = U_m$ (B) $U_e \neq U_m$ (C) $U_e < U_m$ (D) $U_e > U_m$

Open solution

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Q99. JEE Main 2021 (Online) 24th February Evening Shift

Open solution

Match List - I with List - II.

	List I		List II
(a)	Source of microwave frequency	(i)	Radioactive decay of nucleus
(b)	Source of infrared frequency	(ii)	Magnetron
(c)	Source of Gamma Rays	(iii)	Inner shell electrons
(d)	Source of X-rays	(iv)	Vibration of atoms and molecules
		(v)	LASER
		(vi)	RC circuit

Choose the correct answer from the options given below :

(A) (a)-(vi), (b)-(v), (c)-(i), (d)-(iv)

(B) (a)-(ii), (b)-(iv), (c)-(i), (d)-(iii)

(C) (a)-(ii), (b)-(iv), (c)-(vi), (d)-(iii)

(D) (a)-(vi), (b)-(iv), (c)-(i), (d)-(v)

Q100. JEE Main 2020 (Online) 6th September Evening Slot

Open solution

For a plane electromagnetic wave, the magnetic field at a point x and time t is

$\vec{B}(x,t) = \left[1.2 \times 10^{-7} \sin\left(0.5 \times 10^{3}x + 1.5 \times 10^{11}t\right)\hat{k}\right]$ T

The instantaneous electric field $\vec{E}$ corresponding to $\vec{B}$ is :
(speed of light c = 3×10^8 ms^{-1})

(A) $\vec{E}(x,t) = \left[36 \sin\left(1 \times 10^{3}x + 1.5 \times 10^{11}t\right)\hat{i}\right] \frac{V}{m}$

(B) $\vec{E}(x,t) = \left[36 \sin\left(0.5 \times 10^{3}x + 1.5 \times 10^{11}t\right)\hat{k}\right] \frac{V}{m}$

(C) $\vec{E}(x,t) = \left[36 \sin\left(1 \times 10^{3}x + 0.5 \times 10^{11}t\right)\hat{j}\right] \frac{V}{m}$

(D) $\vec{E}(x,t) = \left[-36 \sin\left(0.5 \times 10^{3}x + 1.5 \times 10^{11}t\right)\hat{j}\right] \frac{V}{m}$

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Q101. JEE Main 2020 (Online) 5th September Evening Slot

The correct match between the entries in column I and column II are :

Open solution

I	II
Radiation	**Wavelength**
(a) Microwave	(i) 100 m
(b) Gamma rays	(ii) 10^{-15} m
(c) A.M. radio waves	(iii) 10^{-10} m
(d) X-rays	(iv) 10^{-3} m

(A) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)
(B) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
(C) (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)
(D) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

Q102. JEE Main 2020 (Online) 5th September Morning Slot

An electron is constrained to move along the y-axis with a speed of 0.1 c (c is the speed of light) in the presence of electromagnetic wave, whose electric field is
$\vec{E} = 30\hat{j}\sin\left(1.5 \times 10^{7}t - 5 \times 10^{-2}x\right)$ V/m.
The maximum magnetic force experienced by the electron will be :
(given c = 3×10^{8} ms^{-1} and electron charge = 1.6×10^{-19} C)

Open solution

(A) 4.8×10^{-19} N **(B)** 2.4×10^{-18} N **(C)** 3.2×10^{-18} N **(D)** 1.6×10^{-18} N

Q103. JEE Main 2020 (Online) 4th September Evening Slot

The electric field of a plane electromagnetic wave is given by
$\vec{E} = E_0\left(\hat{x} + \hat{y}\right)\sin\left(kz - \omega t\right)$
Its magnetic field will be given by :

Open solution

(A) $\frac{E_0}{c}\left(\hat{x} + \hat{y}\right)\sin\left(kz - \omega t\right)$
(B) $\frac{E_0}{c}\left(\hat{x} - \hat{y}\right)\sin\left(kz - \omega t\right)$
(C) $\frac{E_0}{c}\left(\hat{x} - \hat{y}\right)\cos\left(kz - \omega t\right)$
(D) $\frac{E_0}{c}\left(-\hat{x} + \hat{y}\right)\sin\left(kz - \omega t\right)$

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Q104. JEE Main 2020 (Online) 4th September Morning Slot

Open solution

Choose the correct option relating wave lengths of different parts of electromagnetic wave spectrum:

(A) $\lambda_{\text{radio waves}} > \lambda_{\text{micro waves}} > \lambda_{\text{visible}} > \lambda_{\text{x-rays}}$

(B) $\lambda_{\text{visible}} > \lambda_{\text{x-rays}} > \lambda_{\text{radio waves}} > \lambda_{\text{micro waves}}$

(C) $\lambda_{\text{visible}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{x-rays}}$

(D) $\lambda_{\text{x-rays}} < \lambda_{\text{micro waves}} < \lambda_{\text{radio waves}} < \lambda_{\text{visible}}$

Q105. JEE Main 2020 (Online) 3rd September Evening Slot

Open solution

The electric field of a plane electromagnetic wave propagating along the x direction in vacuum is $\overrightarrow{E} = E_0\hat{j}\cos(\omega t - kx)$.

The magnetic field $\overrightarrow{B}$, at the moment t = 0 is :

(A) $\overrightarrow{B} = \frac{E_0}{\sqrt{\mu_0 \in_0}}\cos(kx)\hat{j}$

(B) $\overrightarrow{B} = \frac{E_0}{\sqrt{\mu_0 \in_0}}\cos(kx)\hat{k}$

(C) $\overrightarrow{B} = E_0\sqrt{\mu_0 \in_0}\cos(kx)\hat{k}$

(D) $\overrightarrow{B} = E_0\sqrt{\mu_0 \in_0}\cos(kx)\hat{j}$

Q106. JEE Main 2020 (Online) 3rd September Morning Slot

Open solution

The magnetic field of a plane electromagnetic wave is $\overrightarrow{B} = 3 \times 10^{-8}\sin[200\pi(y+ct)]\hat{i}$ T

where c = 3 × 10^8 ms^{-1} is the speed of light. The corresponding electric field is :

(A) $\overrightarrow{E} = -10^{-6}\sin[200\pi(y+ct)]\hat{k}$ V/m

(B) $\overrightarrow{E} = -9\sin[200\pi(y+ct)]\hat{k}$ V/m

(C) $\overrightarrow{E} = 9\sin[200\pi(y+ct)]\hat{k}$ V/m

(D) $\overrightarrow{E} = 3 \times 10^{-8}\sin[200\pi(y+ct)]\hat{k}$

Q107. JEE Main 2020 (Online) 2nd September Evening Slot

Open solution

In a plane electromagnetic wave, the directions of electric field and magnetic field are represented by $\hat{k}$ and $2\hat{i} - 2\hat{j}$, respectively. What is the unit vector along direction of propagation of the wave?

(A) $\frac{1}{\sqrt{5}}\left(\hat{i} + 2\hat{j}\right)$

(B) $\frac{1}{\sqrt{5}}\left(2\hat{i} + \hat{j}\right)$

(C) $\frac{1}{\sqrt{2}}\left(\hat{i} + \hat{j}\right)$

(D) $\frac{1}{\sqrt{2}}\left(\hat{j} + \hat{k}\right)$

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Q108. JEE Main 2020 (Online) 2nd September Morning Slot

Open solution

A plane electromagnetic wave, has
frequency of 2.0×10^{10} Hz and its energy density is 1.02×10^{-8} J/m^3 in vacuum. The amplitude of the magnetic field of the wave is close to
($\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9 \frac{Nm^2}{C^2}$ and speed of light
= 3×10^8 ms^{-1})

(A) 190 nT (B) 150 nT (C) 160 nT (D) 180 nT

Q109. JEE Main 2020 (Online) 9th January Evening Slot

Open solution

A plane electromagnetic wave is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$, with its polarization along the direction $\hat{k}$. The correct form of the magnetic field of the wave would be (here B_0 is an appropriate constant) :

(A) $B_0\frac{\hat{i}-\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(B) $B_0\frac{\hat{i}+\hat{j}}{\sqrt{2}}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(C) $B_0\frac{\hat{j}-\hat{i}}{\sqrt{2}}\cos\left(\omega t + k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

(D) $B_0\hat{k}\cos\left(\omega t - k\frac{\hat{i}+\hat{j}}{\sqrt{2}}\right)$

Q110. JEE Main 2020 (Online) 9th January Morning Slot

Open solution

The electric fields of two plane electromagnetic plane waves in vacuum are given by
$\overrightarrow{E_1} = E_0\hat{j}\cos(\omega t - kx)$ and
$\overrightarrow{E_2} = E_0\hat{k}\cos(\omega t - ky)$
At t = 0, a particle of charge q is at origin with
a velocity $\overrightarrow{v} = 0.8c\hat{j}$ (c is the speed of light in vacuum). The instantaneous force experienced by the particle is :

(A) $E_0q\left(0.8\hat{i} - \hat{j} + 0.4\hat{k}\right)$ (B) $E_0q\left(-0.8\hat{i} + \hat{j} + \hat{k}\right)$

(C) $E_0q\left(0.8\hat{i} + \hat{j} + 0.2\hat{k}\right)$ (D) $E_0q\left(0.4\hat{i} - 3\hat{j} + 0.8\hat{k}\right)$

Q111. JEE Main 2020 (Online) 8th January Evening Slot

Open solution

A plane electromagnetic wave of frequency 25 GHz is propagating in vacuum along the z-direction. At a particular point in space and time, the magnetic field is given by $\overrightarrow{B} = 5 \times 10^{-8}\hat{j}T$. The corresponding electric field $\overrightarrow{E}$ is (speed of light c = 3×10^8 ms^{-1})

(A) 15 $\hat{i}$V / m (B) -15 $\hat{i}$V / m

(C) 1.66×10^{-16} $\hat{i}$V / m (D) -1.66×10^{-16} $\hat{i}$V / m

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Q112. JEE Main 2020 (Online) 7th January Evening Slot

Open solution

The electric field of a plane electromagnetic wave is given by
$\overrightarrow{E} = E_0 \frac{\hat{i}+\hat{j}}{\sqrt{2}} \cos(kz + \omega t)$

At t = 0, a positively charged particle is at the point (x, y, z) = $\left(0, 0, \frac{\pi}{k}\right)$.
If its instantaneous velocity at (t = 0) is $v_0\hat{k}$, the force acting on it due to the wave is :

(A) parallel to $\hat{k}$

(B) parallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(C) antiparallel to $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$

(D) zero

Q113. JEE Main 2020 (Online) 7th January Morning Slot

Open solution

If the magnetic field in a plane electromagnetic wave is given by
$\overrightarrow{B} = 3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j}$ T, then what will be expression for electric field ?

(A) $\overrightarrow{E} = (9\sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{k}$ V/m)

(B) $\overrightarrow{E} = (60\sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{k}$ V/m)

(C) $\overrightarrow{E} = (3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{i}$ V/m)

(D) $\overrightarrow{E} = (3 \times 10^{-8} \sin(1.6 \times 10^3 x + 48 \times 10^{10} t)\hat{j}$ V/m)

Q114. JEE Main 2019 (Online) 12th April Evening Slot

Open solution

A plane electromagnetic wave having a frequency v = 23.9 GHz propagates along the positive z-direction in free space. The peak value of the Electric Field is 60 V/m. Which among the following is the acceptable magnetic field component in the electromagnetic wave ?

(A) $\overrightarrow{B} = 2 \times 10^{-7} \sin(1.5 \times 10^2 x + 0.5 \times 10^{11} t)\hat{j}$

(B) $\overrightarrow{B} = 60 \sin(0.5 \times 10^3 x + 0.5 \times 10^{11} t)\hat{k}$

(C) $\overrightarrow{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z + 1.5 \times 10^{11} t)\hat{i}$

(D) $\overrightarrow{B} = 2 \times 10^{-7} \sin(0.5 \times 10^3 z - 1.5 \times 10^{11} t)\hat{i}$

Q115. JEE Main 2019 (Online) 12th April Morning Slot

Open solution

An electromagnetic wave is represented by the electric field $\overrightarrow{E} = E_0\hat{n}\sin[\omega t + (6y - 8z)]$. Taking unit vectors in x, y and z directions to be $\hat{i}, \hat{j}, \hat{k}$, the direction of propagation $\hat{s}$, is :

(A) $\hat{s} = \frac{3\hat{i}-4\hat{j}}{5}$

(B) $\hat{s} = \frac{-4\hat{k}+3\hat{j}}{5}$

(C) $\hat{s} = \left(\frac{-3\hat{j}+4\hat{k}}{5}\right)$

(D) $\hat{s} = \frac{4\hat{j}-3\hat{k}}{5}$

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Q116. JEE Main 2019 (Online) 10th April Evening Slot

Open solution

Light is incident normally on a completely absorbing surface with an energy flux of 25 W cm^{-2}. If the surface has an area of 25 cm^2, the momentum transferred to the surface in 40 min time duration will be :

(A) 6.3×10^{-4} Ns (B) 5.0×10^{-3} Ns (C) 1.4×10^{-6} Ns (D) 3.5×10^{-6} Ns

Q117. JEE Main 2019 (Online) 10th April Morning Slot

Open solution

The electric field of a plane electromagnetic wave is given by

$\overrightarrow{E} = E_0\hat{i}\cos(kz)cos(\omega t)$

The corresponding magnetic field $\overrightarrow{B}$ is then given by

(A) $\overrightarrow{B} = \frac{E_0}{C}\hat{j}\sin(kz)\sin(\omega t)$ (B) $\overrightarrow{B} = \frac{E_0}{C}\hat{j}\sin(kz)\cos(\omega t)$

(C) $\overrightarrow{B} = \frac{E_0}{C}\hat{j}\cos(kz)\sin(\omega t)$ (D) $\overrightarrow{B} = \frac{E_0}{C}\hat{k}\sin(kz)\cos(\omega t)$

Q118. JEE Main 2019 (Online) 9th April Evening Slot

Open solution

50 W/m^2 energy density of sunlight is normally incident on the surface of a solar panel. Some part of incident energy (25%) is reflected from the surface and the rest is absorbed. The force exerted on 1m^2 surface area will be close to (c = 3×10^8 m/s) :-

(A) 20×10^{-8} N (B) 35×10^{-8} N (C) 10×10^{-8} N (D) 15×10^{-8} N

Q119. JEE Main 2019 (Online) 9th April Morning Slot

Open solution

The magnetic field of a plane electromagnetic wave is given by :

$$\overline{B} = B_0\hat{i}\left[\cos(kz - \omega t)\right] + B_i\hat{j}\cos(kz + \omega t)$$

$B_0 = 3 \times 10^{-5}$ T and B1 = 2×10^{-6} T.

The rms value of the force experienced by a stationary charge Q = 10^{-4} C at z = 0 is closest to :

(A) 0.6 N (B) 0.9 N (C) 3×10^{-2} N (D) 0.1 N

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Q120. JEE Main 2019 (Online) 8th April Evening Slot

The magnetic field of an electromagnetic wave is given by :-

$$\overrightarrow{B} = 1.6 \times 10^{-6} \cos\left(2 \times 10^{7} z + 6 \times 10^{15} t\right) \left(2\hat{i} + \hat{j}\right) \frac{Wb}{m^2}$$

Open solution

The associated electric field will be :-

(A) $\overrightarrow{E} = 4.8 \times 10^{2} \cos\left(2 \times 10^{7} z - 6 \times 10^{15} t\right) \left(-2\hat{i} + \hat{j}\right) \frac{V}{m}$

(B) $\overrightarrow{E} = 4.8 \times 10^{2} \cos\left(2 \times 10^{7} z - 6 \times 10^{15} t\right) \left(2\hat{i} + \hat{j}\right) \frac{V}{m}$

(C) $\overrightarrow{E} = 4.8 \times 10^{2} \cos\left(2 \times 10^{7} z + 6 \times 10^{15} t\right) \left(\hat{i} - 2\hat{j}\right) \frac{V}{m}$

(D) $\overrightarrow{E} = 4.8 \times 10^{2} \cos\left(2 \times 10^{7} z + 6 \times 10^{15} t\right) \left(-\hat{i} + 2\hat{j}\right) \frac{V}{m}$

Q121. JEE Main 2019 (Online) 8th April Morning Slot

A plane electromagnetic wave travels in free space along the x-direction. The electric field component of the wave at a particular point of space and time is E = 6 V m^{-1} along y-direction. Its corresponding magnetic field component, B would be :

Open solution

(A) 2×10^{-8} T along y-direction

(B) 6×10^{-8} T along z-direction

(C) 2×10^{-8} T along z-direction

(D) 6×10^{-8} T along x-direction

Q122. JEE Main 2019 (Online) 12th January Evening Slot

The mean intensity of radiation on the surface of the Sun is about 10^8 W/m^2 . The rms value of the corresponding magnetic field is closet to :

Open solution

(A) 10^2 T

(B) 10^{-2} T

(C) 10^{-4} T

(D) 1 T

Q123. JEE Main 2019 (Online) 12th January Morning Slot

A light wave is incident normally on a glass slab of refractive index 1.5. If 4 % of light gets reflected and the amplitude of the electric field of the incident light is 30 V/m, then the amplitude of the electric field for the wave propagating in the glass medium will be :

Open solution

(A) 6 V/m

(B) 10 V/m

(C) 30 V/m

(D) 24 V/m

Q124. JEE Main 2019 (Online) 11th January Evening Slot

A 27 mW laser beam has a cross-sectional area of 10 mm^2. The magnitude of the maximum electric field in this electromagnetic wave is given by :

[Given permittivity of space $\in_0 = 9 \times 10^{-12}$ SI units, Speed of light c = 3×10^8 m/s]

Open solution

(A) 2 kV/m

(B) 1 kV/m

(C) 1.4 kV/m

(D) 0.7 kV/m

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Q125. JEE Main 2019 (Online) 11th January Morning Slot

An electromagnetic wave of intensity 50 Wm^{-2} enters in a medium of refractive index 'n' without any loss. The ratio of the magnitudes of electric, and the ratio of the magnitudes of magnetic fields of the wave before and after entering into the medium are respectively, given by:

Open solution

(A) $\left(\frac{1}{\sqrt{n}}, \frac{1}{\sqrt{n}}\right)$ (B) $(\sqrt{n}, \sqrt{n})$

(C) $\left(\sqrt{n}, \frac{1}{\sqrt{n}}\right)$ (D) $\left(\frac{1}{\sqrt{n}}, \sqrt{n}\right)$

Q126. JEE Main 2019 (Online) 10th January Evening Slot

The electric field of a plane polarized electromagnetic wave in free space at time t = 0 is given by an expression $\vec{E}(x, y) = 10\hat{j}\cos[(6x + 8z)]$. The magnetic field $\vec{B}$(x,z, t) is given by $-$ (c is the velocity of light)

Open solution

(A) $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

(B) $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x + 8z - 10ct)]$

(C) $\frac{1}{c}\left(6\hat{k} + 8\hat{i}\right)\cos[(6x - 8z + 10ct)]$

(D) $\frac{1}{c}\left(6\hat{k} - 8\hat{i}\right)\cos[(6x - 8z + 10ct)]$

Q127. JEE Main 2019 (Online) 10th January Morning Slot

If the magnetic field of a plane electromagnetic wave is given by (the speed of light = 3×10^8 m/s) B = 100×10^{-6} sin $\left[2\pi \times 2 \times 10^{15}\left(t - \frac{x}{c}\right)\right]$ then the maximum electric field associated with it is -

Open solution

(A) 4.5×10^4 N/C (B) 4×10^4 N/C (C) 6×10^4 N/C (D) 3×10^4 N/C

Q128. JEE Main 2019 (Online) 9th January Evening Slot

The energy associated with electric field is (U_E) and with magnetic field is (U_B) for an electromagnetic wave in free space. Then :

Open solution

(A) $U_E = \frac{U_B}{2}$ (B) $U_E > U_B$ (C) $U_E < U_B$ (D) $U_E = U_B$

Q129. JEE Main 2019 (Online) 9th January Morning Slot

A plane electromagnetic wave of frequency 50 MHz travels in free space along the positive x-direction. At a particular point in space and time, $\vec{E} = 6.3\hat{j}\,V/m$. The corresponding magnetic field $\vec{B}$, at that point will be :

Open solution

(A) $18.9 \times 10^{-8}\,\hat{k}$T (B) $2.1 \times 10^{-8}\,\hat{k}$T

(C) $6.3 \times 10^{-8}\,\hat{k}$T (D) $18.9 \times 10^{8}\,\hat{k}$T

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Q130. JEE Main 2018 (Online) 16th April Morning Slot

A plane electromagnetic wave of wavelength λ has an intensity I. It is propagatting along the positive Y-direction. The allowed expressions for the electric and magnetic fields are givn by :

Open solution

(A)

$$\vec{E} = \sqrt{\frac{2I}{\epsilon_0 C}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right]\hat{k};$$

$$\vec{B} = +\frac{1}{c}E\hat{i}$$

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(B)

$$\vec{E} = \sqrt{\frac{2I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y + ct)\right]\hat{k};$$

$$\vec{B} = \frac{1}{c}E\hat{i}$$

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(C)

$$\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right]\hat{k};$$

$$\vec{B} = \frac{1}{c}E\hat{i}$$

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(D)

$$\vec{E} = \sqrt{\frac{I}{\epsilon_0 c}} \cos\left[\frac{2\pi}{\lambda}(y - ct)\right]\hat{i};$$

$$\vec{B} = \frac{1}{c}E\hat{k}$$

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Q131. JEE Main 2018 (Offline)

Open solution

An EM wave from air enters a medium. The electric fields are

$\overrightarrow{E_1} = E_{01}\hat{x}\cos\left[2\pi v\left(\frac{z}{c}-t\right)\right]$ in air and

$\overrightarrow{E_2} = E_{02}\hat{x}\cos\left[k\left(2z-ct\right)\right]$ in medium,

where the wave number k and frequency ν refer to their values in air. The medium is non-magnetic. If ε_{r_1} and ε_{r_2} refer to relative permittivities of air and medium respectively, which of the following options is correct ?

(A) $\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = 4$

(B) $\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = 2$

(C) $\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = \frac{1}{4}$

(D) $\frac{\varepsilon_{r_1}}{\varepsilon_{r_2}} = \frac{1}{2}$

Q132. JEE Main 2018 (Online) 15th April Evening Slot

Open solution

A plane polarized monochromatic EM wave is traveling in vacuum along z direction such that at t = t_1 it is found that the electric field is zero at a spatial point z_1. The next zero that occurs in its neighbourhood is at z_2. The frequency of the electromagnetic wave is :

(A) $\frac{3\times10^8}{|z_2-z_1|}$

(B) $\frac{1.5\times10^8}{|z_2-z_1|}$

(C) $\frac{6\times10^8}{|z_2-z_1|}$

(D) $\frac{1}{t_1+\frac{|z_2-z_1|}{3\times10^8}}$

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Q133. JEE Main 2018 (Online) 15th April Morning Slot

Open solution

A monochromatic beam of light has a frequency $v = \frac{3}{2\pi} \times 10^{12} Hz$ and is propagating along the direction $\frac{\hat{i}+\hat{j}}{\sqrt{2}}$.

It is polarized along the $\hat{k}$ direction. The acceptable form for the magnetic field is :

(A)

$$\frac{E_o}{C}\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\vec{r}-\left(3\times10^{12}\right)t\right]$$

(B)

$$\frac{E_o}{C}\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}\cos\left[10^4\frac{\left(\hat{i}-\hat{j}\right)}{\sqrt{2}}.\vec{r}-\left(3\times10^{12}\right)t\right]$$

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(C)

$$\frac{E_o}{C}\hat{k}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\vec{r}+\left(3\times10^{12}\right)t\right]$$

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(D)

$$\frac{E_o}{C}\frac{\left(\hat{i}+\hat{j}+\hat{k}\right)}{\sqrt{3}}\cos\left[10^4\frac{\left(\hat{i}+\hat{j}\right)}{\sqrt{2}}.\vec{r}+\left(3\times10^{12}\right)t\right]$$

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Q134. JEE Main 2017 (Online) 9th April Morning Slot

Open solution

The electric field component of a monochromatic radiation is given by

$\vec{E}$ = 2 E_0 $\hat{i}$ cos kz cos ωt

Its magnetic field $\vec{B}$ is then given by :

(A) $\frac{2E_0}{c}$ $\hat{j}$ sin kz cos ωt

(B) $-\frac{2E_0}{c}$ $\hat{j}$ sin kz sin ωt

(C) $\frac{2E_0}{c}$ $\hat{j}$ sin kz sin ωt

(D) $\frac{2E_0}{c}$ $\hat{j}$ cos kz cos ωt

Q135. JEE Main 2017 (Online) 8th April Morning Slot

Open solution

Magnetic field in a plane electromagnetic wave is given by

$\vec{B}$ = B_0 sin (k x + ωt) $\hat{j}T$

Expression for corresponding electric field will be :
Where c is speed of light.

(A) $\vec{E}$ = B_0 c sin (k x + ωt) $\hat{k}$ V/m

(B) $\vec{E}$ = $\frac{B_0}{c}$ sin (k x + ωt) $\hat{k}$ V/m

(C) $\vec{E}$ = $-$ B_0 c sin (kx +ωt) $\hat{k}$ V/m

(D) $\vec{E}$ = B_0 c sin (kx $-\omega$t) $\hat{k}$ V/m

Q136. JEE Main 2016 (Online) 10th April Morning Slot

Open solution

Consider an electromagnetic wave propagating in vacuum. Choose the correct statement :

(A) For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x,t)\left(\hat{y} - \hat{z}\right)$

and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x,t)\left(\hat{y} + \hat{z}\right)$

(B) For an electromagnetic wave propagating in +x direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(y,z,t)\left(\hat{y} + \hat{z}\right)$

and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(y,z,t)\left(\hat{y} + \hat{z}\right)$

(C) For an electromagnetic wave propagating in + y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x,t)\hat{y}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_{yz}(x,t)\hat{z}$

(D) For an electromagnetic wave propagating in + y direction the electric field is $\vec{E} = \frac{1}{\sqrt{2}} E_{yz}(x,t)\hat{z}$ and the magnetic field is $\vec{B} = \frac{1}{\sqrt{2}} B_z(x,t)\hat{y}$

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Q137. JEE Main 2016 (Online) 9th April Morning Slot

Open solution

Microwave oven acts on the principle of :

(A) transferring electrons from lower to higher energy levels in water molecule

(B) giving rotational energy to water molecules

(C) giving vibrational energy to water molecules

(D) giving translational energy to water molecules

Q138. JEE Main 2016 (Offline)

Open solution

Arrange the following electromagnetic radiations per quantum in the order of increasing energy :

A : Blue light B : Yellow light

C : X-ray D : Radiowave.

(A) C, A, B, D (B) B, A, D, C (C) D, B, A, C (D) A, B, D, C

Q139. JEE Main 2015 (Online) 11th April Morning Slot

Open solution

For plane electromagnetic waves propagating in the z direction, which one of the following combination gives the correct possible direction for $\vec{E}$ and $\vec{B}$ field respectively ?

(A) $\left(\hat{i}+2\hat{j}\right)$ and $\left(2\hat{i}-\hat{j}\right)$

(B) $\left(-2\hat{i}-3\hat{j}\right)$ and $\left(3\hat{i}-2\hat{j}\right)$

(C) $\left(2\hat{i}+3\hat{j}\right)$ and $\left(\hat{i}+2\hat{j}\right)$

(D) $\left(3\hat{i}+4\hat{j}\right)$ and $\left(4\hat{i}-3\hat{j}\right)$

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Q140. JEE Main 2015 (Online) 10th April Morning Slot

An electromagnetic wave travelling in the x-direction has frequency of 2×10^{14} Hz and electric field amplitude of 27 Vm^{-1}. From the options given below, which one describes the magnetic field for this wave ?

Open solution

(A) $\vec{B}(x, t) = (3 \times 10^{-8}T)\,\hat{j}$

$\sin [2\pi (1.5 \times 10^{-8}x - 2 \times 10^{14}t)]$

(B) $\vec{B}(x, t) = (9 \times 10^{-8}T)\,\hat{k}$

$\sin [2\pi (1.5 \times 10^{-6}x - 2 \times 10^{14}t)]$

(C) $\vec{B}(x, t) = (9 \times 10^{-8}T)\,\hat{i}$

$\sin [2\pi (1.5 \times 10^{-8}x - 2 \times 10^{14}t)]$

(D) $\vec{B}(x, t) = (9 \times 10^{-8}T)\,\hat{j}$

$\sin [(1.5 \times 10^{-6}\,x - 2 \times 10^{14}t)]$

Q141. JEE Main 2015 (Offline)

A red LED emits light at 0.1 watt uniformly around it. The amplitude of the electric field of the light at a distance of $1\ m$ from the diode is :

Open solution

(A) $5.48\ V/m$ (B) $7.75\ V/m$ (C) $1.73\ V/m$ (D) $2.45\ V/m$

Q142. JEE Main 2014 (Online) 19th April Morning Slot

If microwaves, X rays, infrared, gamma rays, ultra-violet, radio waves and visible parts of the electromagnetic spectrum are represented by M, X, I, G, U, R and V, the following is the arrangement in ascending order of wavelength :

Open solution

(A) R, M, I, V, U, X and G (B) M, R, V, X, U, G and I

(C) G, X, U, V, I, M and R (D) I, M, R, U, V, X and G

Q143. JEE Main 2014 (Online) 12th April Morning Slot

A lamp emits monochromatic green light uniformly in all directions. The lamp is 3% efficient in converting electrical power to electromagnetic waves and consumes 100 W of power. The amplitude of the electric field associated with the electromagnetic radiation at a distance of 5 m from the lamp will be nearly :

Open solution

(A) 1.34 V/m (B) 2.68 V/m (C) 4.02 V/m (D) 5.36 V/m

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Q144. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

An electromagnetic wave of frequency 1×10^{14} hertz is propagating along z - axis. The amplitude of electric field is 4 V/m. If $\in_0 = 8.8 \times 10^{-12}$ $C^2/N\text{-}m^2$, then average energy density of electric field will be :

(A) 35.2×10^{-10} J/m^3

(B) 35.2×10^{-11} J/m^3

(C) 35.2×10^{-12} J/m^3

(D) 35.2×10^{-13} J/m^3

Q145. JEE Main 2014 (Online) 11th April Morning Slot

Open solution

Match the **List - I** (Phenomenon associated with electromagnetic radiation) with **List - II** (Part of electromagnetic spectrum) and select the correct code from the choices given below the lists :

List - I		**List - II**	
(I)	Doublet of sodium	(A)	Visible radiation
(II)	Wavelength corresponding to temperature associated with the isotropic radiation filling all space	(B)	Microwave
(III)	Wavelength emitted by atomic hydrogen in interstellar space	(C)	Short radiowave
(IV)	Wavelength of radiation arising from two close energy levels in hydrogen	(D)	X - rays

(A) (I)-(A), (II)-(B), (III)-(B), (IV)-(C)

(B) (I)-(A), (II)-(B), (III)-(C), (IV)-(C)

(C) (I)-(D), (II)-(C), (III)-(A), (IV)-(B)

(D) (I)-(B), (II)-(A), (III)-(D), (IV)-(A)

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Q146. JEE Main 2014 (Online) 9th April Morning Slot

Match **List I** (Wavelength range of electromagnetic spectrum) with **List II** (Method of production of these waves) and select the **correct** option from the options given below the lists.

Open solution

List I		List II	
(a)	700 nm to 1 mm	(i)	Vibration of atoms and molecules.
(b)	1 nm to 400 nm	(ii)	Inner shell electrons in atoms moving from one energy level to a lower level.
(c)	$< 10^{-3}$ nm	(iii)	Radioactive decay of the nucleus.
(d)	1 mm to 0.1 m	(iv)	Magnetron valve.

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(A) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(B) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(C) (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

(D) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

Q147. JEE Main 2014 (Offline)

During the propagation of electromagnetic waves in a medium :

Open solution

(A) Electric energy density is double of the magnetic energy density.

(B) Electric energy density is half of the magnetic energy density.

(C) Electric energy density is equal to the magnetic energy density.

(D) Both electric and magnetic energy of densities are zero.

Q148. JEE Main 2014 (Offline)

Match List - 1 (Electromagnetic wave type) with List - 2 (Its association/application) and select the correct option from the choices given below the lists:

Open solution

List 1	List 2
1. Infrared waves	(i) To treat muscular strain
2. Radio waves	(ii) For broadcasting
3. X-rays	(iii) To detect fracture of bones
4. Ultraviolet rays	(iv) Absorbed by the ozone layer of the atmosphere

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(A) $1-iv,\ 2-iii,\ 3-ii,\ 4-i$

(B) $1-i,\ 2-ii,\ 3-iv,\ 4-iii$

(C) $1-iii,\ 2-ii,\ 3-i,\ 4-iv$

(D) $1-i,\ 2-ii,\ 3-iii,\ 4-iv$

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Q149. JEE Main 2013 (Online) 25th April Morning Slot

This question has Statement-1 and Statement-2. Of the four choices given after the Statements, choose the one that best describes the two Statements.
Statement-1: Out of radio waves and microwaves, the radio waves undergo more diffraction.
Statement-2: Radio waves have greater frequency compared to microwaves.

Open solution

(A) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of Statement-1

(B) Statement-1 is false, Statement-2 is true.

(C) Statement-1 is true, Statement-2 is false.

(D) Statement-1 is true, Statement-2 is true but Statement-2 is not the correct explanation of Statement-1

Q150. JEE Main 2013 (Online) 23th April Morning Slot

A parallel plate capacitor of area 60 cm^2 and separation 3 mm is charged initially to $90\mu C$. If the medium between the plate gets slightly conducting and the plate loses the charge initially at the rate of 2.5×10^{-8} C/s, then what is the magnetic field between the plates?

Open solution

(A) 2.5×10^{-8} T (B) 2.0×10^{-7} T (C) 1.63×10^{-11} T (D) Zero

Q151. JEE Main 2013 (Online) 23th April Morning Slot

Select the correct statement from the following :

Open solution

(A) Electromagnetic waves cannot travel in vacuum.

(B) Electromagnetic waves are longitudinal waves.

(C) Electromagnetic waves are produced by charges moving with uniform velocity.

(D) Electromagnetic waves carry both energy and momentum as they propagate through space.

Q152. JEE Main 2013 (Online) 22th April Morning Slot

A plane electromagnetic wave in a non-magnetic dielectric medium is given by $\vec{E} = \vec{E}_0 \sin(4 \times 10^{-7}x - 50t)$ with distance being in meter and time in seconds. The dielectric constant of the medium is :

Open solution

(A) 2.4 (B) 5.8 (C) 8.2 (D) 4.8

Q153. JEE Main 2013 (Online) 9th April Morning Slot

Photons of an electromagnetic radiation has an energy 11 keV each. To which region of electromagnetic spectrum does it belong?

Open solution

(A) X-ray region (B) Ultra violet region

(C) Infrared region (D) Visible region

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Q154. JEE Main 2013 (Offline)

The magnetic field in a travelling electromagnetic wave has a peak value of $20\ nT$. The peak value of electric field strength is :

(A) $3V/m$ (B) $6V/m$ (C) $9V/m$ (D) $12V/m$

Open solution

Q155. AIEEE 2012 (Online) 26th May

A radio transmitter transmits at $830\ \text{kHz}$. At a certain distance from the transmitter magnetic field has amplitude $4.82 \times 10^{-11}\ \text{T}$. The electric field and the wavelength are respectively

(A) $0.014\ \text{N/C},\ 36\ \text{m}$ (B) $0.14\ \text{N/C},\ 36\ \text{m}$

(C) $0.14\ \text{N/C},\ 360\ \text{m}$ (D) $0.014\ \text{N/C},\ 360\ \text{m}$

Open solution

Q156. AIEEE 2012 (Online) 26th May

The frequency of X-rays; γ-rays and ultraviolet rays are respectively a, b and c then

(A) $a < b;\ b > c$ (B) $a > b;\ b > c$ (C) $a < b < c$ (D) $a = b = c$

Open solution

Q157. AIEEE 2012 (Online) 19th May

An electromagnetic wave with frequency ω and wavelength λ travels in the $+y$ direction. Its magnetic field is along $+x$-axis. The vector equation for the associated electric field (of amplitude E_0) is

(A) $\vec{E} = -E_0 \cos\left(\omega t + \frac{2\pi}{\lambda} y\right)\hat{x}$ (B) $\vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right)\hat{x}$

(C) $\vec{E} = E_0 \cos\left(\omega t - \frac{2\pi}{\lambda} y\right)\hat{z}$ (D) $\vec{E} = -E_0 \cos\left(\omega t + \frac{2\pi}{\lambda} y\right)\hat{z}$

Open solution

Q158. AIEEE 2012

An electromagnetic wave in vacuum has the electric and magnetic field $\vec{E}$ and $\vec{B}$, which are always perpendicular to each other. The direction of polarization is given by $\vec{X}$ and that of wave propagation by $\vec{k}$. Then

(A) $\vec{X} \| \vec{B}$ and $\vec{X} \| \vec{B} \times \vec{E}$

(B) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{E} \times \vec{B}$

(C) $\vec{X} \| \vec{B}$ and $\vec{k} \| \vec{E} \times \vec{B}$

(D) $\vec{X} \| \vec{E}$ and $\vec{k} \| \vec{B} \times \vec{E}$

Open solution

Q159. AIEEE 2006

The rms value of the electric field of the light coming from the Sun is $720\ N/C$. The average total energy density of the electromagnetic wave is

(A) $4.58 \times 10^{-6}\ J/m^3$ (B) $6.37 \times 10^{-9}\ J/m^3$

(C) $81.35 \times 10^{-12}\ J/m^3$ (D) $3.3 \times 10^{-3}\ J/m^3$

Open solution

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Q160. AIEEE 2004

An electromagnetic wave of frequency $v = 3.0\ MHz$ passes from vacuum into a dielectric medium with permittivity $\in = 4.0$. Then

(A) wave length is halved and frequency remains unchanged

(B) wave length is doubled and the frequency becomes half

(C) wave length is doubled and the frequency remains unchanged

(D) wave length and frequency both remain unchanged.

Open solution

Q161. AIEEE 2004

A radiation of energy E falls normally on a perfectly reflecting surface. The momentum transferred to the surface is

(A) Ec (B) $2E/c$ (C) E/c (D) E/c^2

Open solution

Q162. AIEEE 2002

Infrared radiation is detected by

(A) spectrometer (B) pyrometer (C) nanometer (D) photometer

Open solution

Q163. AIEEE 2002

Electromagnetic waves are transverse in nature is evident by

(A) polarization (B) interference (C) reflection (D) diffraction

Open solution

Numerical 22 questions

Q164. JEE Main 2026 (Online) 23rd January Morning Shift

The equation of the electric field of an electromagnetic wave propagating through free space is given by : $E = \sqrt{377}\sin\left(6.27 \times 10^3 t - 2.09 \times 10^{-5} x\right)\mathrm{N/C}$

The average power of the electromagnetic wave is $\left(\frac{1}{\alpha}\right)\mathrm{W/m^2}$. The value of α is ____ .

$\left(\text{Take } \sqrt{\frac{\mu_0}{\varepsilon_o}} = 377 \text{ in SI units}\right)$

Open solution

Q165. JEE Main 2026 (Online) 22nd January Morning Shift

The electric field of a plane electromagnetic wave, travelling in an unknown nonmagnetic medium is given by,

$E_y = 20\sin\left(3 \times 10^6 x - 4.5 \times 10^{14}\mathrm{t}\right)\mathrm{V/m}$

(where x, t and other values have S.I. units). The dielectric constant of the medium is ____

(speed of light in free space is $3 \times 10^8\ \mathrm{m/s}$)

Open solution

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Q166. JEE Main 2026 (Online) 21st January Evening Shift

An electromagnetic wave of frequency 100 MHz propagates through a medium of conductivity, $\sigma = 10\,\text{mho/m}$. The ratio of maximum conduction current density to maximum displacement current density is ____.

$\left[\text{Take } \frac{1}{4\pi\epsilon_0} = 9 \times 10^9\,\text{Nm}^2/\text{C}^2\right]$

Open solution

Q167. JEE Main 2025 (Online) 4th April Evening Shift

If an optical medium possesses a relative permeability of $\frac{10}{\pi}$ and relative permittivity of $\frac{1}{0.0885}$, then the velocity of light is greater in vacuum than that in this medium by _____ times.

$\left(\mu_0 = 4\pi \times 10^{-7}\text{H/m}, \epsilon_0 = 8.85 \times 10^{-12}\,\text{F/m}, \text{c} = 3 \times 10^8\,\text{m/s}\right)$

Open solution

Q168. JEE Main 2025 (Online) 23rd January Evening Shift

A time varying potential difference is applied between the plates of a parallel plate capacitor of capacitance $2.5\mu\,\text{F}$. The dielectric constant of the medium between the capacitor plates is 1 . It produces an instantaneous displacement current of 0.25 mA in the intervening space between the capacitor plates, the magnitude of the rate of change of the potential difference will be _____ Vs^{-1}.

Open solution

Q169. JEE Main 2025 (Online) 22nd January Evening Shift

A parallel plate capacitor of area $A = 16\,\text{cm}^2$ and separation between the plates 10 cm , is charged by a DC current. Consider a hypothetical plane surface of area $\text{A}_0 = 3.2\,\text{cm}^2$ inside the capacitor and parallel to the plates. At an instant, the current through the circuit is 6A. At the same instant the displacement current through A_0 is _____ mA .

Open solution

Q170. JEE Main 2023 (Online) 31st January Morning Shift

In a medium the speed of light wave decreases to 0.2 times to its speed in free space. The ratio of relative permittivity to the refractive index of the medium is $x : 1$. The value of x is _____.

(Given speed of light in free space $= 3 \times 10^8\,\text{m s}^{-1}$ and for the given medium $\mu_\text{r} = 1$)

Open solution

Q171. JEE Main 2023 (Online) 30th January Morning Shift

A point source of light is placed at the centre of curvature of a hemispherical surface. The source emits a power of $24\,\text{W}$. The radius of curvature of hemisphere is $10\,\text{cm}$ and the inner surface is completely reflecting. The force on the hemisphere due to the light falling on it is _____ $\times 10^{-8}\,\text{N}$.

Open solution

Q172. JEE Main 2022 (Online) 29th July Evening Shift

Nearly 10% of the power of a $110\,\text{W}$ light bulb is converted to visible radiation. The change in average intensities of visible radiation, at a distance of $1\,\text{m}$ from the bulb to a distance of $5\,\text{m}$ is $a \times 10^{-2}\,\text{W/m}^2$. The value of 'a' will be _____.

Open solution

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Q173. JEE Main 2022 (Online) 29th June Evening Shift

The displacement current of 4.425 μA is developed in the space between the plates of parallel plate capacitor when voltage is changing at a rate of 10^6 Vs^{-1}. The area of each plate of the capacitor is 40 cm^2. The distance between each plate of the capacitor is x $\times$ 10^{-3} m. The value of x is _____.

(Permittivity of free space, ε_0 = 8.85 $\times$ 10^{-12} C^2 N^{-1} m^{-2}).

Open solution

Q174. JEE Main 2022 (Online) 29th June Morning Shift

The intensity of the light from a bulb incident on a surface is 0.22 W/m^2. The amplitude of the magnetic field in this light-wave is _____ $\times$ 10^{-9} T.

(Given : Permittivity of vacuum $\in_0$ = 8.85 $\times$ 10^{-12} C^2 N^{-1}-m^{-2}, speed of light in vacuum c = 3 $\times$ 10^8 ms^{-1})

Open solution

Q175. JEE Main 2021 (Online) 31st August Morning Shift

The electric field in an electromagnetic wave is given by E = (50 NC^{-1}) sinω (t $-$ x/c)

The energy contained in a cylinder of volume V is 5.5 $\times$ 10^{-12} J. The value of V is _____ cm^3. (given $\in_0$ = 8.8 $\times$ $10^{-12}C^2N^{-1}m^{-2}$)

Open solution

Q176. JEE Main 2021 (Online) 27th August Evening Shift

A plane electromagnetic wave with frequency of 30 MHz travels in free space. At particular point in space and time, electric field is 6 V/m. The magnetic field at this point will be x $\times$ 10^{-8} T. The value of x is _____.

Open solution

Q177. JEE Main 2021 (Online) 26th August Morning Shift

The electric field in a plane electromagnetic wave is given by

$\vec{E} = 200\cos\left[\left(\frac{0.5\times10^3}{m}\right)x - \left(1.5\times10^{11}\,\frac{rad}{s}\times t\right)\right]\frac{V}{m}\hat{j}$. If this wave falls normally on a perfectly reflecting surface having an area of 100 cm^2. If the radiation pressure exerted by the E.M. wave on the surface during a 10 minute exposure is $\frac{x}{10^9}\frac{N}{m^2}$. Find the value of x .

Open solution

Q178. JEE Main 2021 (Online) 17th March Evening Shift

The electric field intensity produced by the radiation coming from a 100 W bulb at a distance of 3 m is E. The electric field intensity produced by the radiation coming from 60W at the same distance is $\sqrt{\frac{x}{5}}$E. Where the value of x = _____.

Open solution

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Q179. JEE Main 2021 (Online) 17th March Evening Shift

Seawater at a frequency f = 9×10^2 Hz, has permittivity $\varepsilon = 80\varepsilon_0$ and resistivity $\rho = 0.25\ \Omega$m. Imagine a parallel plate capacitor is immersed in seawater and is driven by an alternating voltage source V(t) = $V_0 \sin(2\pi ft)$. Then the conduction current density becomes 10^x times the displacement current density after time t = $\frac{1}{800}$s. The value of x is _____. (Given : $\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9$ Nm^2C^{-2})

Open solution

Q180. JEE Main 2021 (Online) 17th March Morning Shift

If 2.5×10^{-6} N average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is _____ W/cm^2. (Round off to the Nearest Integer)

(Assume complete absorption and normal incidence conditions are there)

Open solution

Q181. JEE Main 2021 (Online) 26th February Morning Shift

A radiation is emitted by 1000W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is $x \times 10^{-1}$ V/m. Value of x is _____. (Rounded off to the nearest integer) [Take $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2N^{-1} m^{-2}, c = 3×10^8 ms^{-1}]

Open solution

Q182. JEE Main 2021 (Online) 25th February Evening Shift

The peak electric field produced by the radiation coming from the 8W bulb at a distance of 10 m is $\frac{x}{10}\sqrt{\frac{\mu_0 c}{\pi}}\frac{V}{m}$. The efficiency of the bulb is 10% and it is a point source. The value of x is _____.

Open solution

Q183. JEE Main 2021 (Online) 24th February Evening Shift

An electromagnetic wave of frequency 3 GHz enters a dielectric medium of relative electric permittivity 2.25 from vacuum. The wavelength of this wave in that medium will be _____ $\times 10^{-2}$ cm.

Open solution

Q184. JEE Main 2021 (Online) 24th February Morning Shift

An electromagnetic wave of frequency 5 GHz, is travelling in a medium whose relative electric permittivity and relative magnetic permeability both are 2. Its velocity in this medium is _____ $\times 10^7$ m/s.

Open solution

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Q185. JEE Main 2020 (Online) 6th September Morning Slot

Suppose that intensity of a laser is $\frac{315}{\pi}$ W/m^2.
The rms electric field, in units of V/m associated with this source is close to the nearest integer is _____.

Open solution

$\in_0 = 8.86 \times 10^{-12}$ C^2 Nm^{-2}; c = 3×10^8 ms^{-1})

MCQ (More than One Correct Answer) 1 questions

Q186. AIEEE 2002

Which of the following are not electromagnetic waves?

(A) cosmic rays (B) gamma rays (C) β-rays (D) X-rays

Open solution

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Answer Key

MCQ (Single Correct Answer)

1. **B**	2. **B**	3. **B**	4. **B**	5. **A**
6. **B**	7. **A**	8. **A**	9. **C**	10. **C**
11. **C**	12. **B**	13. **A**	14. **A**	15. **B**
16. **C**	17. **C**	18. **C**	19. **B**	20. **C**
21. **B**	22. **C**	23. **D**	24. **A**	25. **B**
26. **D**	27. **A**	28. **C**	29. **B**	30. **D**
31. **A**	32. **A**	33. **B**	34. **B**	35. **B**
36. **D**	37. **C**	38. **B**	39. **D**	40. **A**
41. **A**	42. **B**	43. **B**	44. **C**	45. **A**
46. **C**	47. **D**	48. **C**	49. **C**	50. **C**
51. **A**	52. **D**	53. **A**	54. **A**	55. **D**
56. **C**	57. **A**	58. **A**	59. **B**	60. **D**
61. **C**	62. **C**	63. **C**	64. **A**	65. **C**
66. **B**	67. **D**	68. **A**	69. **B**	70. **C**
71. **D**	72. **C**	73. **A**	74. **B**	75. **B**
76. **B**	77. **C**	78. **A**	79. **B**	80. **C**
81. **D**	82. **A**	83. **B**	84. **B**	85. **C**
86. **A**	87. **C**	88. **C**	89. **D**	90. **C**
91. **D**	92. **A**	93. **D**	94. **C**	95. **A**
96. **C**	97. **B**	98. **A**	99. **B**	100. **D**
101. **B**	102. **A**	103. **D**	104. **A**	105. **C**
106. **B**	107. **C**	108. **C**	109. **A**	110. **C**
111. **A**	112. **C**	113. **A**	114. **D**	115. **C**
116. **B**	117. **A**	118. **A**	119. **A**	120. **D**
121. **C**	122. **C**	123. **D**	124. **C**	125. **C**
126. **B**	127. **D**	128. **D**	129. **B**	130. **A**
131. **C**	132. **B**	133. **A**	134. **C**	135. **A**
136. **A**	137. **B**	138. **C**	139. **B**	140. **B**
141. **D**	142. **C**	143. **B**	144. **C**	145. **B**
146. **D**	147. **C**	148. **D**	149. **C**	150. **D**

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151. **D**	152. **B**	153. **A**	154. **B**	155. **D**
156. **A**	157. **C**	158. **B**	159. **A**	160. **A**
161. **B**	162. **B**	163. **A**		

Numerical

164. **2**	165. **4**	166. **1800**	167. **6**	168. **100**
169. **1200**	170. **5**	171. **4**	172. **84**	173. **8**
174. **43**	175. **500**	176. **2**	177. **354**	178. **3**
179. **6**	180. **25**	181. **137**	182. **2**	183. **667**
184. **15**	185. **194**			

MCQ (More than One Correct Answer)

186. **A, C**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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